Laser Filamentation: Nonlinear Self-Channeling of Light

PHY204 - Project P# 17

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Abstract. We model femtosecond laser filamentation in fused silica at a carrier wavelength of 800 nm using a cylindrically symmetric generalised nonlinear Schrödinger equation coupled to a Drude plasma rate equation, integrated by a Strang split-step scheme that combines spectral group-velocity dispersion, Crank–Nicolson radial diffraction, and pointwise nonlinear updates. Without parameter tuning, the simulation reproduces the canonical filamentation signatures: Kerr-driven self-focusing up to the Marburger collapse distance, plasma-mediated intensity clamping at $\sim 10^{13}$ W/cm^2, asymmetric pulse splitting consistent with the dynamic-replenishment picture of Mlejnek *et al.*, and an angular spectrum that aligns with the analytic X-wave hyperbolae of normal-GVD conical waves to within $\sim 1\%$. The clamped intensity, peak free-electron density, filament length and per-millimetre energy loss all fall within the bands compiled by Couairon and Kolesik for fused silica at 800 nm. A toggle-by-toggle sensitivity analysis identifies the minimal physics required to produce a filament — diffraction, dispersion, the Kerr response, and at least one of multiphoton absorption or plasma defocusing — and shows that avalanche ionisation, recombination and inverse-bremsstrahlung absorption are quantitatively negligible at our working point.

1 Introduction

Laser filamentation is the long-distance propagation of an intense ultrashort laser pulse in a transparent nonlinear medium with little transverse spreading. The phenomenon was first reported by Braun *et al.* [1], who observed self-channelling of femtosecond pulses over more than 20 m in air with a narrow channel whose energy, diameter and spectrum remained approximately constant along the propagation. Although that experiment was performed in air, the same Kerr-versus-plasma mechanism operates in condensed media, with different material parameters and a much shorter filament length — millimetres rather than metres. Such behaviour is unexpected from the standpoint of linear optics, where a beam of finite waist diffracts over its Rayleigh range and cannot remain spatially localised over many such ranges.

The underlying mechanism is intensity-dependent: an intense pulse modifies the refractive index of the medium it propagates through. In a centrosymmetric dielectric the macroscopic polarisation admits the perturbative expansion

$$\mathbf{P} = \varepsilon_0\big(\chi^{(1)}\mathbf{E} + \chi^{(3)}|\mathbf{E}|^2\mathbf{E} + \cdots\big), \tag{1}$$

in which even-order terms vanish by inversion symmetry and the cubic, intensity-dependent $\chi^{(3)}$ contribution dresses the linear susceptibility. Combined with $n^2 = 1 + \chi_{\text{eff}}$ and truncated at leading order in the intensity, Eq. (1) yields the Kerr approximation

$$n = n_0 + n_2\, I, \tag{2}$$

where $I = |E|^2$ is the optical intensity and $n_2 \propto \chi^{(3)}/n_0$ is the Kerr coefficient. For $n_2 > 0$, the on-axis refractive index exceeds its off-axis value: wavefronts accumulate more optical phase on axis than at the edges of the beam, so the wavefront curves inward and the beam contracts toward the high-index core. This is optical Kerr self-focusing.

At sufficiently high intensity the pulse ionises the medium and creates free electrons. The free-electron contribution to the refractive index is negative, so the plasma acts as a defocusing lens and counteracts Kerr self-focusing. A filament emerges from the dynamical balance between Kerr focusing, diffraction, multiphoton absorption and plasma defocusing.

We restrict ourselves to the single-filament, cylindrically symmetric regime, in which the field is a function of three coordinates only: the transverse radius r, the retarded time τ in the moving frame of the pulse, and the propagation distance z. The goal is not to address multi-filamentation or transverse symmetry breaking, but to understand the evolution of a single intense pulse in this (r, τ, z) representation. The spatio-temporal description is essential because each temporal slice of the pulse self-focuses at a different rate, and the plasma generated by the leading edge influences the propagation of the trailing edge — an effect that purely spatial models cannot capture.

This project has three main objectives:

1. First, identify the minimal physical ingredients needed to reproduce filamentation;
2. Second, implement a stable numerical method for the propagation equation;
3. Finally, interpret the simulated evolution in terms of self-focusing, plasma defocusing and dynamic replenishment.

All simulations reported here are performed for fused silica at the carrier wavelength $\lambda_0 = 800$ nm, taken as a representative condensed medium. The corresponding material parameters follow the practitioner-guide compilation of Couairon and Kolesik [6] and are tabulated in Section 3. The physical model and the numerical strategy build directly on prior work: Braun *et al.* [1] provide the experimental benchmark, Mlejnek *et al.* [2,3] establish the generalised nonlinear Schrödinger equation that we adopt, and the preliminary project notebook on the advection–diffusion equation [7,8] introduced the numerical primitives — Fourier-spectral propagation and Crank–Nicolson finite differences combined through operator splitting — that we adapt here to the paraxial optical case.

The remainder of the paper is organised as follows. Section 2 sets out the theoretical framework, Section 3 the numerical method and its validation, and Section 4 the simulation results and their comparison with the literature. Section 5 summarises our findings and outlines natural extensions. Appendix A derives the propagation equation from Maxwell's equations, Appendix B derives the critical power and Marburger collapse distance used as reference values throughout, and Appendix C lists the contributions of each author.

2 Theoretical background

2.1 Paraxial envelope approximation

We describe the optical field by a slowly varying complex envelope $E(r, z, t)$ multiplying a rapidly oscillating carrier. The physical electric field is therefore written

$$\mathcal{E}(r, z, t) \;=\; \mathrm{Re}\Big[E(r, z, t)\, e^{i(kz-\omega_0 t)}\Big], \tag{3}$$

with E slowly varying on optical scales. The pulse propagates predominantly along z, so the longitudinal variation of E is slow compared with the optical wavelength; this is the slowly-varying-envelope approximation (SVEA). We further restrict ourselves to a cylindrically symmetric field — azimuthally invariant — which excludes the multi-filament regime and any transverse symmetry breaking.

In the retarded frame moving at the group velocity, with retarded time $\tau = t - z/v_g$ where $v_g = \partial\omega/\partial k|_{\omega_0}$ is the group velocity at the carrier frequency, a standard propagation model for femtosecond filamentation is

$$\frac{\partial E}{\partial z} = \frac{i}{2k}\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r}\right)E - \frac{ik''}{2}\frac{\partial^2 E}{\partial \tau^2} - \frac{\sigma}{2}(1 + i\omega_0\tau_c)\rho E - \frac{\beta_K}{2}|E|^{2K-2}E + ik_0 n_2|E|^2 E. \tag{4}$$

We adopt the convention that $k_0 = 2\pi/\lambda_0$ is the *vacuum* wavenumber and $k = n_0 k_0$ the wavenumber in the unperturbed medium; the Kerr coefficient $k_0 n_2$ in Eq. (4) is therefore equivalent to $(\omega_0/c)\, n_2$. The right-hand side of Eq. (4) collects the five physical effects implemented in our solver: transverse diffraction, group-velocity dispersion, the full Drude plasma response (both absorption and defocusing), multiphoton absorption, and the instantaneous Kerr self-action. Two additional contributions discussed in the literature — the delayed Raman response written as a convolution in time [3] and self-steepening — are not included here because they introduce both physical and numerical complexity beyond the scope of a first-version solver; we return to them as priority extensions in Section 5.

Each term in Eq. (4) is summarised in Table 1. The associated material constants are: $k'' = \partial^2 k/\partial\omega^2$, the group-velocity-dispersion coefficient; $\rho(r, z, t)$, the free-electron density; σ, the inverse-bremsstrahlung cross-section; τ_c, the electron collision time; β_K, the K-photon absorption coefficient; and n_2, the Kerr nonlinear index.

Table 1. Physical interpretation of the terms in the propagation equation.

Term	Physical effect
$\frac{i}{2k}\left(\partial_r^2 + r^{-1}\partial_r\right) E$	Transverse diffraction. It tends to broaden the beam in r.
$-\frac{ik''}{2}\partial_\tau^2 E$	Group-velocity dispersion. It reshapes the temporal profile.
$-\frac{\sigma}{2}\rho E$	Plasma absorption through inverse bremsstrahlung.
$-\frac{i\sigma\omega_0\tau_c}{2}\rho E$	Plasma-induced phase shift, equivalent to a negative refractive-index contribution.
$-\frac{\beta_K}{2}\lvert E\rvert^{2K-2}E$	Multiphoton absorption, important only at high intensity.
$ik_0 n_2 \lvert E\rvert^2 E$	Kerr self-focusing/self-phase modulation.

2.2 Electron-density model

The laser field does not merely propagate through the medium; it also modifies it by liberating bound electrons. This back-action is described by the free-electron density $\rho(r,\tau,z)$, which couples to the local intensity $I = |E|^2$ through a Drude-type rate equation. Following Mlejnek *et al.* [2] and Couairon & Kolesik [6] we write

$$\frac{\partial \rho}{\partial \tau} = \underbrace{\frac{\beta_K}{K\hbar\omega_0}\,|E|^{2K}\left(1-\frac{\rho}{\rho_{nt}}\right)}_{\text{multiphoton ionisation}} + \underbrace{\frac{\sigma}{U_i}\,\rho\,|E|^2}_{\text{avalanche}} - \underbrace{a\,\rho^2}_{\text{recombination}}, \tag{5}$$

where σ is the inverse-bremsstrahlung cross-section, U_i is the ionisation potential of the medium, β_K is the K-photon absorption coefficient, $K = \lceil U_i/\hbar\omega_0 \rceil$ is the minimum number of photons required to bridge the gap, ρ_{nt} is the neutral-molecule density, and a is the radiative-recombination coefficient. The MPI rate coefficient σ_K is related to β_K by the photon-number balance

$$\sigma_K = \frac{\beta_K}{K\hbar\omega_0\,\rho_{nt}}, \tag{6}$$

so that the energy $\beta_K|E|^{2K}$ removed from the field by MPA per unit volume and time is exactly the energy $K\hbar\omega_0$ deposited into each of the $\sigma_K|E|^{2K}\rho_{nt}$ ionisation events occurring per unit volume and time.

The first term on the right-hand side describes *multiphoton ionisation* (MPI). A bound electron absorbs K photons simultaneously to overcome the ionisation potential $U_i < K\hbar\omega_0$. Since each ionisation event requires the energy $K\hbar\omega_0$ and removes power $\beta_K|E|^{2K}$ from the field, the resulting electron production rate is $\beta_K|E|^{2K}/(K\hbar\omega_0)$. The very steep $|E|^{2K}$ scaling (with $K = 5$ in fused silica at 800 nm) makes MPI negligible at low intensity but explosive near the self-focused peak: it is the trigger that switches plasma on once the Kerr lens has driven the intensity above the MPI threshold. The factor $(1-\rho/\rho_{nt})$ is the *neutral-depletion* or *Pauli-blocking* factor: it counts the fraction of bound electrons still available for ionisation, suppresses the source as ρ approaches the neutral density, and guarantees $\rho \le \rho_{nt}$. In our regime $\rho_{\text{peak}}/\rho_{nt} \sim 10^{-4}$, so this factor is numerically close to unity and the depletion correction is negligible; it nevertheless becomes essential once ionisation approaches saturation, e.g. at higher input powers or in gases with smaller neutral density.

The second term describes *avalanche* (or cascade) ionisation. Free electrons already present in the medium are accelerated by the laser field through inverse Bremsstrahlung, i.e. they absorb energy from the optical field during collisions with neutral atoms or molecules. They can then gain enough kinetic energy to ionise neighbouring bound electrons upon impact, thereby producing new free electrons and seeding further ionisation events. The rate is therefore proportional both to the existing density ρ and to the intensity $|E|^2$: without seed electrons it cannot start, and the per-electron heating rate scales linearly with the laser intensity. The factor σ/U_i converts the absorbed power per electron, $\sigma|E|^2$, into a production rate by dividing by the energy needed to liberate one new electron.

The third term, $-a\rho^2$, accounts for radiative *recombination* of free electrons with positive ions. The quadratic dependence is natural when ion and electron densities are equal, since each recombination event requires one electron *and* one ion. It removes free carriers from the channel, but its characteristic timescale ($\sim 1/(a\rho_{\text{peak}})$) is much longer than the pulse duration τ_p in our regime, so it is essentially inactive during the pulse and only matters for the long-lived plasma track left behind.

In the parameter regime considered here — fused silica, clamped on-axis intensity $\sim 10^{13}$ W/cm^2, peak free-electron density $\rho_{\text{peak}} \sim 5\times10^{18}$/cm^3, pulse duration ~ 100 fs — multiphoton ionisation dominates the avalanche contribution by roughly an order of magnitude and recombination is negligible during the pulse. Avalanche becomes the dominant ionisation channel at higher input powers and longer pulse

durations, while recombination governs the relaxation of the ionisation track on nanosecond timescales after the pulse has passed.

2.3 Critical power and qualitative filament picture

The threshold for filamentation onset can be understood by balancing diffraction against Kerr self-focusing. The Kerr term (2) makes the refractive index slightly larger on axis than at the edges of the beam, so an intense beam writes a positive lens into the medium it traverses. Below a critical power P_{cr} this lens is too weak to overcome diffraction and the beam spreads as in linear optics; above it, Kerr focusing prevails and the beam contracts toward catastrophic on-axis collapse. For a continuous-wave beam the threshold scales as

$$P_{\mathrm{cr}} \simeq \frac{\lambda_0^2}{2\pi\, n_0\, n_2}, \tag{7}$$

up to numerical factors that depend on the beam profile and on the precise convention used to define P_{cr} (see Appendix B).

In a real femtosecond pulse the collapse does not continue indefinitely. As the intensity rises, multi-photon absorption removes energy from the field and the resulting free electrons lower the local refractive index near the axis, so the plasma acts as a negative lens. The filament is therefore not produced by Kerr self-focusing alone but by the competition between Kerr focusing, diffraction, nonlinear absorption and plasma defocusing. The dynamical balance between these four mechanisms produces the phenomenon known as *intensity clamping*: instead of diverging, the on-axis intensity reaches a quasi-stationary plateau set by the ionisation threshold. Experiments in air report a clamped intensity around 7×10^{13} W/cm^2 with a narrow channel that persists over many Rayleigh ranges [1]. In condensed media at 800 nm, the practitioner-guide compilation of Couairon and Kolesik [6] places the clamped on-axis intensity in the range 10^{13}–10^{14} W/cm^2 and the peak free-electron density in the range 10^{18}–10^{19}/cm^3; both ranges will be recovered by our simulation in Section 4.

The filament should not be pictured as a static self-guided pulse. In the spatio-temporal (r, τ, z) description, the leading edge of the pulse focuses first, generates plasma, and is subsequently defocused or absorbed; energy still present in the rear and outer regions of the pulse can refocus later and form a new high-intensity peak on axis. The observed filament therefore corresponds to a *sequence* of focusing, plasma-generation and replenishment events rather than to a single unchanged object propagating forward [2,3].

An equivalent and complementary description, developed by Faccio, Couairon and Di Trapani [4], treats the femtosecond filament not primarily as a Kerr-versus-plasma balance but as a self-organised conical wave-packet — an "X-wave" in normal-dispersion media. In the angular-spectral plane $(k_\perp, \omega)$, the propagation-invariant solutions of the linear paraxial-plus-GVD operator lie on the hyperbolae

$$k_\perp \;=\; \pm\,|\Omega|\,\sqrt{k_0\, k''}, \qquad \Omega = \omega - \omega_0. \tag{8}$$

Once the nonlinearity drives the field above the filamentation threshold, the pulse spontaneously reshapes onto these loci, and the off-axis hyperbolic wings continually feed energy back into the central spot. In the conical-wave picture, plasma plays the role of trigger and clamp rather than that of the channelling mechanism itself. The two pictures are mathematically consistent in our regime; we recover the conical-wave signature in the far-field spectrum in Section 4.4.

3 Numerical method

We integrate the propagation equation (4) coupled to the plasma rate equation (5) on a discretised cylindrical (r, τ, z) grid by a Strang operator-splitting scheme in z. Each operator is treated by a method tailored to its mathematical character: the group-velocity-dispersion operator is integrated spectrally in τ, the radial-diffraction operator implicitly in r by Crank–Nicolson, and the nonlinear and plasma sectors explicitly through pointwise complex exponentials and a forward-Euler march in τ. The strategy is a direct generalisation of the spectral and Crank–Nicolson primitives first practised on the one-dimensional advection–diffusion equation in the project notebook [8].

3.1 Numerical primitives: Fourier-spectral and Crank–Nicolson methods

Before implementing Eq. (4) we studied the one-dimensional advection–diffusion equation

$$\frac{\partial u}{\partial t} + V\,\frac{\partial u}{\partial x} \;=\; D\,\frac{\partial^2 u}{\partial x^2} + f(u), \tag{9}$$

which shares the structural ingredients of the paraxial propagation equation: linear derivative operators (a first-order advection and a second-order diffusion) augmented by a local nonlinear source $f(u)$. The principal difference is that in the optical problem the evolution variable is the propagation distance z rather than the physical time t.

The preliminary notebook introduced two strategies that we re-use throughout this work. The first is the Fourier-spectral method. If

$$u(x,t) = \int \hat{u}(k,t) e^{ikx}\, dk,$$

then spatial derivatives become simple multiplications in Fourier space:

$$\partial_x u \longleftrightarrow ik\hat{u}, \qquad \partial_x^2 u \longleftrightarrow -k^2\hat{u}.$$

For the linear case ($f = 0$), the PDE therefore becomes, for each Fourier mode,

$$\frac{\partial \hat{u}}{\partial t} = -(Dk^2 + iVk)\hat{u}.$$

This is an ordinary differential equation, so over one step Δt, it is straightforward:

$$\hat{u}(k, t+\Delta t) = \hat{u}(k,t) \exp\left[-(Dk^2 + iVk)\Delta t\right]. \tag{10}$$

Numerically the passage between $u(x,t)$ and $\hat{u}(k,t)$ is performed by the Fast Fourier Transform (FFT). The method is efficient because it replaces the discretisation of the derivative operators by a multiplication of each Fourier component by an exact exponential factor, with no truncation error from finite-difference stencils.

The second primitive is the Crank–Nicolson method, a finite-difference scheme that is unconditionally stable and second-order accurate for parabolic partial differential equations. The continuous problem is discretised on regular grids in both space and time,

$$t_n = n\,\Delta t, \quad n = 0, \ldots, N-1, \qquad x_j = j\,\Delta x, \quad j = 0, \ldots, J-1,$$

with $\Delta t = T/N$ and $\Delta x = L/J$. The numerical approximation to the analytic solution at the grid point $(j\Delta x, n\Delta t)$ is denoted U_j^n, and the values at a fixed time level n are collected in the vector $\mathbf{U}^n = (U_0^n, \ldots, U_{J-1}^n)^{\mathsf{T}}$.

Derivatives are approximated by difference quotients between neighbouring nodes. The continuous limit $\partial_x u = \lim_{\Delta x \to 0}[u(x+\Delta x) - u(x)]/\Delta x$ is replaced on the grid by the forward time difference

$$\left.\frac{\partial u}{\partial t}\right|_{j,\,n} \approx \frac{U_j^{n+1} - U_j^n}{\Delta t},$$

i.e. the slope between two consecutive time levels at the same node. The second spatial derivative is obtained by applying the same construction twice, yielding the centred three-point stencil

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{j,\,n} \approx \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{\Delta x^2},$$

which we write compactly as $(L\mathbf{U}^n)_j$ for a tridiagonal matrix L.

The Crank–Nicolson scheme evaluates the right-hand side at the midpoint $t_n + \Delta t/2$ by averaging the values at t_n and t_{n+1}. For a linear operator L the update reads

$$\frac{\mathbf{U}^{n+1} - \mathbf{U}^n}{\Delta t} = \frac{1}{2}\big(L\mathbf{U}^n + L\mathbf{U}^{n+1}\big) + f(\mathbf{U}^n),$$

where the nonlinear source is kept explicit for simplicity. The midpoint centering makes the scheme second-order accurate in Δt. For linear parabolic operators, the resulting implicit averaging gives an amplification factor of modulus at most one for all Δt, which makes the linear Crank–Nicolson scheme unconditionally stable. Rearranging yields a linear system

$$A\,\mathbf{U}^{n+1} \;=\; B\,\mathbf{U}^n + F^n, \qquad F^n \equiv \Delta t\, f(\mathbf{U}^n),$$

which is solved at each step for $\mathbf{U}^{n+1}$. Both A and B are tridiagonal in our application; A is factored once and re-used at every step.

3.2 Operator splitting and overall algorithm

The right-hand side of the propagation equation is separated into a linear part $\mathcal{L}$ and a local nonlinear part $\mathcal{N}$,

$$\frac{\partial E}{\partial z} = \mathcal{L}\,E + \mathcal{N}(E,\rho)\,E, \qquad \mathcal{L} = \frac{i}{2k}\left(\partial_r^2 + \frac{1}{r}\partial_r\right) - \frac{ik''}{2}\partial_\tau^2, \tag{11}$$

where $\mathcal{N}$ collects the Kerr, plasma-defocusing, plasma-absorption and multiphoton-absorption contributions to Eq. (4). Over one longitudinal step of size Δz we use the symmetric (Strang) splitting

$$E(z+\Delta z) \;\approx\; e^{\frac{\Delta z}{2}\mathcal{L}}\, e^{\Delta z\,\mathcal{N}}\, e^{\frac{\Delta z}{2}\mathcal{L}}\, E(z), \tag{12}$$

which is second-order accurate in Δz provided each substep is itself at least second order.

Listing 1. Schematic propagation algorithm.

```
initialise E(r, tau, z=0) as Gaussian beam x Gaussian pulse
initialise rho(r, tau, z=0) = 0
for n in 0 .. N_z - 1:
    E = linear_half_step(E, dz/2) # spectral GVD + Crank-Nicolson diffraction
    rho = update_plasma_density(E, rho) # MPI + avalanche - recombination, Euler in tau
    E = nonlinear_step(E, rho, dz) # Kerr + plasma defocusing + MPA + plasma absorption
    E = linear_half_step(E, dz/2)
    E = E * radial_absorber(r) # damp residual outgoing rings
    record diagnostics every M steps
```

3.3 Linear half-step: spectral GVD and Crank–Nicolson diffraction

The linear operator is integrated using a combination of the Fourier-spectral and Crank–Nicolson primitives introduced in Section 3.1. The two contributions of $\mathcal{L}$ commute and decouple cleanly in the temporal Fourier domain: writing $\hat{E}(r,\Omega,z)$ for the Fourier transform of E along τ, the GVD term becomes a multiplication by $+i(k''/2)\Omega^2$ at every Ω (the sign follows from $\partial_\tau^2 \to -\Omega^2$ in NumPy's $e^{-i\Omega\tau}$ Fourier convention), while diffraction acts on r alone. Each linear half-step of size $\Delta z/2$ therefore consists of four substeps:

1. FFT in τ: $E(r,\tau) \to \hat{E}(r,\Omega)$;
2. apply the GVD phase $\hat{E}(r,\Omega) \to \hat{E}(r,\Omega)\exp\!\big[+\tfrac{ik''}{2}\Omega^2\tfrac{\Delta z}{2}\big]$;
3. inverse FFT: $\hat{E}(r,\Omega) \to E(r,\tau)$;
4. solve the radial Crank–Nicolson update at every τ.

The radial Laplacian is discretised in conservative form on a *cell-centred* grid $r_j = (j+\tfrac{1}{2})\Delta r$, $j = 0,\dots,N_r-1$:

$$\big(\nabla_\perp^2 E\big)_j = \frac{1}{r_j\,\Delta r^2}\Big[r_{j+\frac{1}{2}}(E_{j+1}-E_j) - r_{j-\frac{1}{2}}(E_j - E_{j-1})\Big]. \tag{13}$$

The cell-centred staggering avoids the apparent $1/r$ singularity at the axis: setting $r_{-1/2} = 0$ makes the inner-face flux vanish identically, which is exactly the regularity condition $\partial_r E|_{r=0} = 0$ for a smooth axisymmetric field, and no ghost point is required. At the outer face $r_{N_r-1/2} = R_{\max}$ we rely on the absorbing mask introduced in Section 3.5 to suppress any flux that reaches the wall.

With the discrete Laplacian operator $\mathbf{D}$ defined as the tridiagonal matrix encoding the stencil (13), the Crank–Nicolson update of size $\Delta z/2$ for the diffraction equation $\partial_z E = (i/2k)\,\nabla_\perp^2 E$ reads

$$\Big(\mathbf{I} - \tfrac{i\,\Delta z}{8\,k}\,\mathbf{D}\Big)\,\mathbf{E}^{n+1} = \Big(\mathbf{I} + \tfrac{i\,\Delta z}{8\,k}\,\mathbf{D}\Big)\,\mathbf{E}^{n}, \tag{14}$$

where $\mathbf{E}^n$ is the radial profile at fixed τ. Crank–Nicolson is unconditionally stable and second-order accurate in both Δr and Δz. Since the coefficient matrix is independent of τ and z, the implicit operator is LU-factored once with `scipy.linalg.lu_factor` during initialisation and the factorisation is reused at every linear substep through `scipy.linalg.lu_solve`; the resulting tridiagonal solve, repeated N_τ times per linear half-step, dominates the cost of $\mathcal{L}$.

3.4 Nonlinear step: plasma rate equation and pointwise field update

The nonlinear operator $\mathcal{N}$ is local in r and τ and is therefore evaluated point-wise. Within each step we (i) march the plasma density along τ from the initial condition $\rho(r, \tau = -\tau_{\max}) = 0$ to obtain $\rho(r, \tau)$ at the current z, and (ii) advance the field over the full Δz by a single complex exponential.

The plasma rate equation (5) is integrated by forward Euler in τ, an explicit single-step scheme that is well within its stability window for our parameters ($\Delta\tau \sim 2$ fs against a plasma rise time of ~ 10 fs):

$$\rho(r, \tau+\Delta\tau) = \rho(r, \tau) + \Delta\tau \left[\sigma_K |E|^{2K} (\rho_{nt} - \rho) + \frac{\sigma}{U_i} \rho |E|^2 - a\, \rho^2 \right]. \tag{15}$$

After each τ-substep the result is clipped to $[0, \rho_{nt}]$ to suppress any numerical overshoot near saturation. This clipping is diagnostic only and never triggers in the converged regime. The recombination term $-a\rho^2$ is included for completeness but is negligible during the sub-picosecond pulse.

The field update assembles the four nonlinear contributions into a single complex exponent and applies them in one step over Δz:

$$E \;\to\; E \exp\left[i\, \Delta z \left(\frac{k n_2}{n_0} |E|^2 - \frac{k}{2 n_0 \rho_c} \rho \right) - \Delta z \left(\frac{\beta_K}{2} |E|^{2K-2} + \frac{\sigma}{2} \rho \right) \right]. \tag{16}$$

The two imaginary contributions in the exponent describe Kerr self-phase modulation and plasma defocusing (a negative refractive-index change); the two real contributions describe multiphoton absorption and inverse-bremsstrahlung absorption. Each of the four terms can be toggled independently in the implementation, a feature exploited in Section 3.6 to recover the linear and analytic limits.

3.5 Grids, initial condition and boundary treatment

All three coordinates are sampled on uniform grids:

- *Radial:* $N_r = 384$ cell-centred points on $[0, R_{\max}]$ with $R_{\max} = 400\,\mu$m, giving $\Delta r \approx 1.04\,\mu$m.
- *Retarded-time:* $N_\tau = 512$ points on $[-\tau_{\max}, \tau_{\max}]$ with $\tau_{\max} = 500$ fs, giving $\Delta\tau \approx 1.95$ fs.
- *Longitudinal:* $\Delta z = 3\,\mu$m over $L_z = 12$ mm, i.e. $N_z = 4000$ steps.

The radial sampling is set by the requirement to resolve the few-μm filament core with several grid points. The temporal box is wide enough ($\sim 6\, t_p$) for the pulse to remain numerically zero at the FFT-periodic boundaries throughout the propagation. The longitudinal step is small enough that the per-step nonlinear phase $k_0 n_2 I_{\text{peak}} \Delta z$ remains $\lesssim 0.1$ rad in the clamped state, well within the linearisation limit of the split-step scheme.

The initial condition is a separable Gaussian beam and pulse,

$$E(r, \tau, z{=}0) = E_0 \exp\!\left(-r^2/w_0^2\right) \exp\!\left(-\tau^2/t_p^2\right), \tag{17}$$

with $w_0 = 50$ μm, $t_p = 85$ fs and the peak amplitude $E_0 = \sqrt{I_0}$ fixed by $I_0 = 2P_{\text{in}}/(\pi w_0^2)$ at $P_{\text{in}} = 5\, P_{cr}$.

The radial boundary at $r = 0$ is handled implicitly by the cell-centred grid, as discussed above. At the outer boundary, rather than impose a true outgoing-wave condition, the field is multiplied at the end of every full z-step by a super-Gaussian mask

$$M(r) = \exp\!\left[-\left(\frac{r - r_{\text{abs}}}{R_{\max} - r_{\text{abs}}} \right)^8 \right] \quad \text{for } r > r_{\text{abs}}, \qquad M(r) = 1 \text{ otherwise}, \tag{18}$$

acting only in the outer 8% of the radial box. This damps any wave that has diffracted past the relevant filament region and prevents artificial reflection back toward the axis. The temporal grid is treated as periodic by the FFT, an approximation that is accurate provided the pulse remains numerically zero at $\pm\tau_{\max}$; we have verified that this is the case for the parameters above.

3.6 Validation tests

Before running the full nonlinear simulation we verify each component of the solver against a known analytic limit, using the toggles introduced in Section 3.4 to isolate one effect at a time. Four tests are performed:

(A) *Linear diffraction.* All nonlinear terms and GVD off; the beam half-width should evolve as $w(z) = w_0\sqrt{\ln 2/2}\,\sqrt{1+(z/z_R)^2}$ with $z_R = \pi n_0 w_0^2/\lambda_0$.
(B) *Pure GVD.* Diffraction and nonlinearities off; the on-axis $1/e$-intensity half-width should evolve as $(t_p/\sqrt{2})\sqrt{1+(2zk''/t_p^2)^2}$.
(C) *Subcritical Kerr.* $P_{\text{in}} = 0.5\,P_{cr}$ with plasma and MPA off; the beam should exhibit only mild self-focusing and no collapse, providing a sanity check that the implementation does not generate spurious blow-up.
(D) *Plane-wave MPA.* All terms off except multiphoton absorption, applied to a uniform plane wave; the intensity should follow the closed-form depletion law $I(z) = I_0/[1+(K-1)\beta_K I_0^{K-1} z]^{1/(K-1)}$.

Tests A, B and D reproduce their analytic targets to better than 0.1% relative error on the grids of Section 3.5; Test C confirms the qualitative no-collapse behaviour expected at sub-critical power. The corresponding plots are included in the project notebook accompanying this report.

3.7 Implementation and computational cost

The main numerical difficulty is the wide range of physical scales: the field envelope, plasma density, multiphoton coefficient and step sizes span many orders of magnitude, so direct SI-unit calculations risk overflow in expressions of the form $|E|^{2K}$ or underflow in small coefficients. Two safeguards are used to control this. First, ρ is clipped to $[0, \rho_{nt}]$ — a diagnostic safeguard rather than a physical model, used only to detect unstable parameter choices that violate the saturation bound. Second, convergence tests in Δz, N_r and N_τ confirm that the qualitative features are grid-independent. A separate concern is the radial boundary: the filament core is narrow but the surrounding energy reservoir is physically important, so $R_{\max}$ is chosen large enough to preserve the reservoir while keeping the simulation cost manageable, and the super-Gaussian absorber damps any flux that does reach the wall.

The solver is implemented in Python 3 as a single `Filament` class mirroring the structure of the `Spectral` and `CrankNicolson` classes used in the advection–diffusion notebook [8]. The `__init__` method builds the grids and pre-factors the Crank–Nicolson operator; `set_initial_gaussian` applies Eq. (17); `step` executes one longitudinal step via the routines above; and a `propagate` driver advances the field through N_z steps and records scalar and snapshot diagnostics on a coarser sub-sampled grid so that the memory footprint is independent of N_z. Only NumPy and SciPy are required (`lu_factor`, `lu_solve`, the FFT, and the Bessel-J_0 special function used in the Hankel transform of Section 4.4); no domain-specific filamentation library is used. The full physics run with the parameters listed in Section 3.5 completes in approximately five minutes on a laptop CPU, which makes parameter scans practical.

4 Results and discussion

This section presents the simulation results obtained for the silica parameters listed in Section 3.5: a Gaussian pulse with $w_0 = 50\,\mu$m, $t_p = 85$ fs, and $\lambda_0 = 800$ nm. The presentation is organised in three telescoping levels of detail. We begin with a sub-critical reference run that establishes the baseline below the filamentation threshold (Section 4.1). We then present the canonical filament at the working power $P_{\text{in}} = 5\,P_{cr}$, first through on-axis scalar diagnostics and the temporal pulse-splitting signature (Section 4.2), then through the full spatio-temporal structure of the filament core and plasma channel (Section 4.3), and finally through the angular-spectral X-wave signature (Section 4.4). The section closes with a toggle-by-toggle sensitivity analysis that isolates the role of each physical ingredient (Section 4.5) and a discussion of the model's limitations and a quantitative comparison with the literature (Section 4.6).

4.1 Reference case: sub-critical propagation

Before presenting the filamentation regime we propagate the model at an input power well below the self-focusing threshold, $P_{\text{in}} = 0.5\,P_{cr}$, with all nonlinear and plasma terms enabled. This sub-critical run serves as a negative control: it tests whether the solver, when driven below threshold, does anything qualitatively different from a bounded, well-behaved linear beam. Linear diffraction and dispersion in isolation are checked separately by Tests A and B of Section 3.6; here we additionally verify that the Kerr, plasma and absorption terms produce no filament-like behaviour at sub-critical intensity.

Figure 1 shows the on-axis peak intensity $I_{\text{peak}}(z)$ and the beam half-width $w(z)$ over the full propagation, with the linear-diffraction prediction $w(z) = w_0\sqrt{\ln 2/2}\,\sqrt{1+(z/z_R)^2}$ overlaid for reference. Two features are notable. First, the peak intensity grows by only 4.1% above its input value

$I_0 \simeq 2.8 \times 10^{10}$ W/cm^2 over the full 6 mm — far short of the multi-decade rise characteristic of catastrophic Kerr collapse. Second, the beam HWHM *decreases* mildly with z, from 29.4 μm at the input to 28.3 μm at the output, while the linear prediction *increases* to 32 μm. The Kerr nonlinearity is therefore active even at sub-critical power and partially cancels diffraction, but the residual self-focusing is bounded: with $P_{\rm in} < P_{cr}$ the Kerr lens cannot overcome diffraction globally, and the beam settles into a slow sub-Rayleigh contraction rather than a true collapse.

Plasma generation is negligible throughout. The peak free-electron density never exceeds $\rho_{\rm peak} \sim 3 \times 10^6$/cm^3, twelve orders of magnitude below the values reached in Section 4.2 and sixteen orders below the neutral density ρ_{nt}. Multiphoton ionisation, scaling as I^K with $K = 5$, is simply too weak at $I \sim 10^{10}$ W/cm^2 to seed any plasma response, and the total energy drop from MPA and inverse-bremsstrahlung over the entire run is below the numerical noise floor ($\lesssim 10^{-4}$).

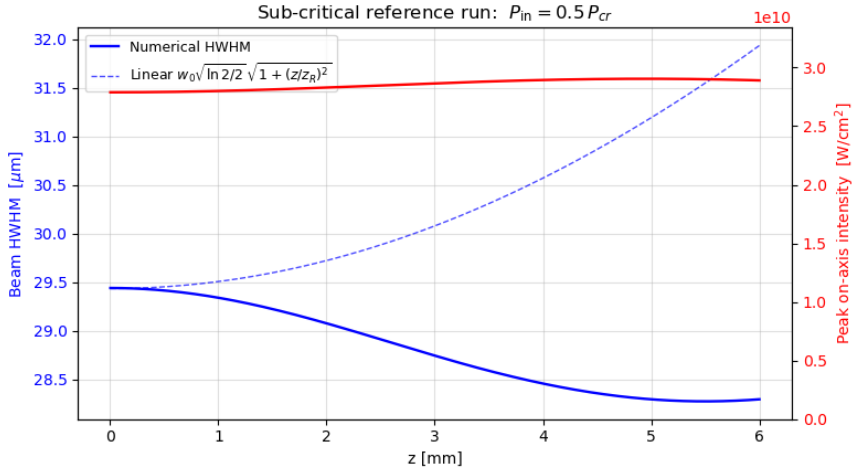


Fig. 1. Sub-critical reference run, $P_{\rm in} = 0.5\,P_{cr}$, full V1.5 model (Kerr, plasma, MPA, avalanche, recombination and plasma absorption all enabled). *Blue, left axis:* beam half-width $w(z)$ from the simulation (solid) and the linear-diffraction prediction $w_0\sqrt{\ln 2/2}\,\sqrt{1+(z/z_R)^2}$ (dashed). The numerical curve sits below the linear curve by $\sim$4 μm at $z = L_z$, indicating mild bounded self-focusing rather than divergence. *Red, right axis:* peak on-axis intensity $I_{\rm peak}(z)$, which grows by only 4% over the full propagation. The peak plasma density (not plotted) stays below 3×10^6/cm^3 everywhere — twelve orders of magnitude below the levels reached in Sec. 4.2. No filament forms below P_{cr}.

This sub-critical baseline plays two roles in what follows. First, it demonstrates that the dramatic dynamics of subsequent sections are not artefacts of the solver but require $P_{\rm in}$ above the threshold defined by Eq. (7). Second, the bounded $I_{\rm peak}(z)$ and the slowly contracting $w(z)$ here provide a clean reference against which the clamped, oscillating $I_{\rm peak}(z)$ and the few-μm collapsed filament core of the next section can be read directly.

4.2 Canonical filament at $P_{\rm in} = 5\,P_{cr}$

We propagate the model at the working power $P_{\rm in} = 5\,P_{cr}$, corresponding to a peak input intensity $I_0 = 2.79 \times 10^{11}$ W/cm^2, over the propagation length $L_z = 12$ mm so that the entire filament life-cycle — build-up, clamping and decay back to a diffracting beam — is contained within the simulation box. All other parameters are those of Section 3.5. The results are summarised in two figures: an on-axis scalar-diagnostic triple in Fig. 2 and a two-dimensional propagation map in Fig. 3.

Self-focusing, plasma onset, clamping and decay. Figure 2 shows the longitudinal evolution of the peak on-axis intensity $I_{\rm peak}(z)$, the peak free-electron density $\rho_{\rm peak}(z)$, and the beam half-width $w(z)$. Four regimes can be identified.

(i) Kerr build-up, $z \lesssim z_{\rm Marburger} = 3.80$ mm. Only Kerr self-focusing is active: $I_{\rm peak}$ grows monotonically by more than an order of magnitude while w contracts smoothly from 29.4 μm to a few micrometres, and the plasma density remains at the numerical floor.

(ii) Plasma ignition, $z \approx z_{\mathrm{Marburger}}$. The on-axis intensity reaches $\sim 10^{13}$ W/cm^2, at which point multiphoton ionisation ($\propto I^K$ with $K = 5$) takes over abruptly: ρ_{peak} rises by approximately seven orders of magnitude over a fraction of a millimetre and reaches 6.5×10^{18}/cm^3, well below the critical Drude density $\rho_c = 1.74 \times 10^{21}$/cm^3 ($\rho_{\mathrm{peak}}/\rho_c \simeq 4 \times 10^{-3}$). The field therefore never approaches optical breakdown.

(iii) Plateau and refocusing oscillations, $3.5 \lesssim z \lesssim 9$ *mm.* The plasma response defocuses the on-axis field and I_{peak} *clamps* near 1.1×10^{13} W/cm^2 (maximum 1.13×10^{13} W/cm^2 at $z = 4.37$ mm). Within this plateau, $I_{\mathrm{peak}}(z)$ is modulated by 18 distinct local maxima with a typical spacing of $\sim 200\,\mu$m: these are the canonical Kerr-versus-plasma refocusing oscillations of a dynamically balanced filament. The filament core, defined operationally by $w(z) < 10\,\mu$m, persists from $z \approx 3$ mm to $z \approx 12$ mm and reaches a minimum HWHM of $2.55\,\mu$m at $z = 4.67$ mm.

(iv) Decay, $z \gtrsim 9$ *mm.* The on-axis intensity decays monotonically by more than an order of magnitude as cumulative multiphoton and inverse-bremsstrahlung absorption deplete the on-axis energy faster than the off-axis reservoir can replenish it. By $z = L_z$ the on-axis intensity has fallen back to 4.7×10^{11} W/cm^2 — a factor of 24 below the clamped value and within a factor of two of the input — and the beam half-width has re-expanded to $10.2\,\mu$m. The filament has fully died: the V1.5 model in this parameter regime sustains a single filament over $\sim$6 mm and no longer.

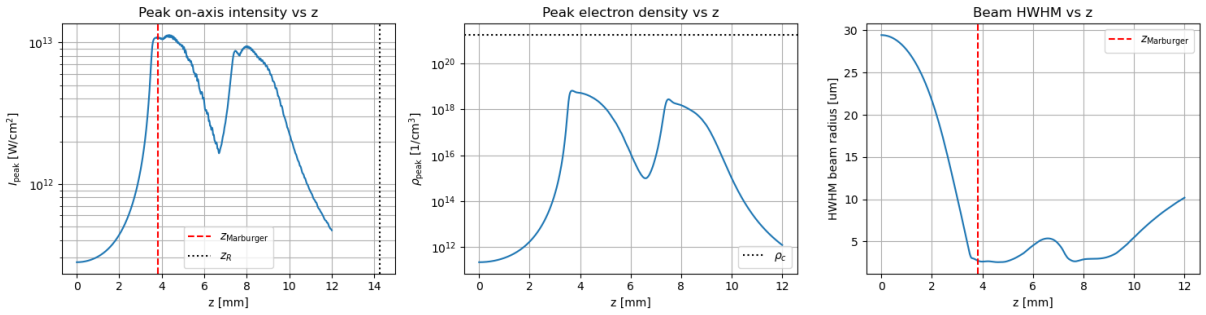


Fig. 2. Canonical filament at $P_{\mathrm{in}} = 5\,P_{cr}$, full V1.5 model, $L_z = 12$ mm. *Left:* peak on-axis intensity $I_{\mathrm{peak}}(z)$ (log scale), with the Marburger collapse position $z_{\mathrm{Marburger}} = 3.80$ mm and the linear Rayleigh range $z_R = 14.3$ mm marked. The plateau between $z \approx 3.5$ mm and $\sim$9 mm is modulated by Kerr-vs-plasma refocusing oscillations of period $\sim 200\,\mu$m; past the plateau the intensity decays back toward the input level. *Centre:* peak free-electron density $\rho_{\mathrm{peak}}(z)$ reaches 6.5×10^{18}/cm^3 at $z \sim 4$ mm, well below the critical density ρ_c, then decays once the on-axis intensity drops below the MPI threshold. *Right:* beam half-width $w(z)$ contracts to a $\sim 2.6\,\mu$m filament core near $z_{\mathrm{Marburger}}$ and re-expands to $\sim 10\,\mu$m by the end of the box. The full life-cycle of the filament — build-up, plateau, decay — is now contained within the simulation.

Pulse splitting and dynamic replenishment. The on-axis propagation map of Fig. 3 resolves the temporal structure that the scalar diagnostics integrate over. Up to $z \sim z_{\mathrm{Marburger}}$ the on-axis field appears as a single bright stripe centred on $\tau = 0$, brightening slowly as the beam self-focuses; at $z \approx 2$ mm the leading-side and trailing-side peaks are still degenerate (ratio 1.00). At and just past $z_{\mathrm{Marburger}}$ the stripe forks into two trails separated by an intensity dip that grows from zero to nearly an order of magnitude over the following millimetre. The relative brightness of the two trails subsequently *evolves* along z, and that evolution is itself the spatio-temporal fingerprint of dynamic replenishment.

Immediately past the fork (at $z = 4$ mm) the leading peak is fractionally brighter than the trailing one (ratio 1.04): both halves of the pulse have just collapsed and plasma effects have not yet had time to discriminate between them. From $z \sim 5$ to $\sim$9 mm the leading peak is preferentially drained by the cumulative absorption that it has experienced, while the trailing peak strengthens as energy from the off-axis reservoir refocuses onto axis; the ratio falls below unity (e.g. 0.78 at $z = 8$ mm). Past $z \approx 9$ mm the trailing peak in turn is depleted while the leading peak — replenished from the outer halo — recovers, and by $z = 10$ mm the leading peak dominates again with a ratio of 1.55. This non-monotonic exchange between the two halves of the pulse corresponds exactly to the dynamic-replenishment scenario introduced by Mlejnek *et al.* [2]: the filament does not propagate as a single unchanged object but as a succession of focusing, plasma-mediated decay, and trailing- or leading-edge replenishment events. Time-symmetric splitting (e.g. Kerr plus GVD with no plasma) would generate two trails of equal brightness throughout, which we do not observe.

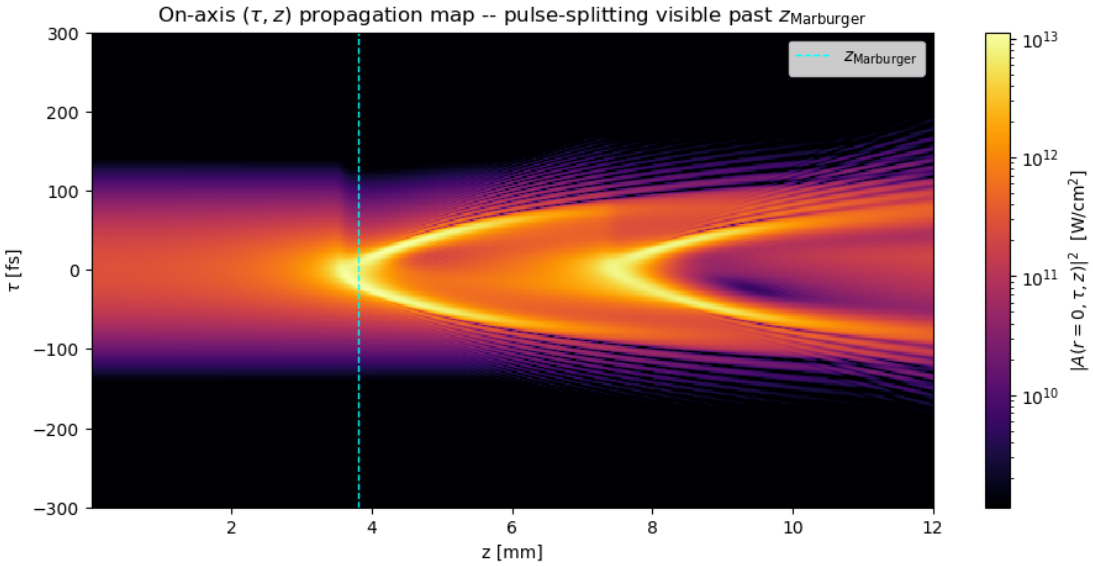


Fig. 3. On-axis (τ, z) propagation map of $|A(r=0,\tau,z)|^2$ at $P_{\mathrm{in}} = 5\,P_{cr}$, $L_z = 12$ mm, on a logarithmic colour scale. The cyan dashed line marks $z_{\mathrm{Marburger}}$. The pulse splits into a leading-side trail (negative τ) and a trailing-side trail (positive τ) at and past the collapse. The relative brightness of the two trails evolves non-monotonically: leading-dominant immediately past the fork, trailing-dominant in the central plateau region (driven by plasma defocusing of the leading edge and refocusing of the trailing edge), and leading-dominant again past $z \sim 9$ mm as the trailing peak in turn is depleted. By $z = L_z$ both trails have faded back to within a factor of two of the input intensity.

Quantitative comparison with literature. The clamped on-axis intensity $\sim 10^{13}$ W/cm^2 and the peak plasma density $\sim 6 \times 10^{18}$/cm^3 both fall within the bands reported by Couairon and Kolesik for fused silica at 800 nm (10^{13}–10^{14} W/cm^2 and 10^{18}–10^{19}/cm^3, respectively) [6]. The filament length, defined as the z-range over which I_{peak} exceeds half its maximum, is $\Delta z = 5.7$ mm, squarely in the millimetre-scale range expected for filamentation in bulk silica. The total energy drop of 17.6% from MPA and inverse-bremsstrahlung absorption over the full 12 mm corresponds to approximately 1.5%/mm during the filament plateau, also consistent with the values quoted in the practitioner-guide compilation [6]. The V1.5 model therefore produces a single filament whose on-axis scalar and temporal signatures lie within the literature range and exhibit the canonical dynamic-replenishment fingerprint, before any further analysis of its spatial or angular-spectral structure.

4.3 Spatial structure: filament core and plasma channel

The on-axis diagnostics of Section 4.2 integrate over the radial structure of the pulse and so cannot resolve the geometric extent of the filament. To complete the picture we examine the full two-dimensional intensity field $|E(r,\tau)|^2$ and the plasma density $\rho(r,\tau)$ at twelve snapshots taken every millimetre across the propagation, $z = 1, 2, \ldots, 12$ mm. The four phases of the filament life-cycle — build-up, plasma ignition, plateau with refocusing oscillations, and final decay — are then captured in a single coherent sequence.

Near-field intensity. Figure 4 shows $|E(r,\tau)|^2$ denoted as $|A(r,\tau)|^2$ in the figure at all twelve snapshots on a logarithmic colour scale, with the radial axis mirrored to $-r$ to emphasise the cylindrical symmetry. The four phases identified in Section 4.2 are clearly visible.

Build-up (panels a–c, $z = 1$–3 mm): the pulse remains smooth and approximately Gaussian, contracting under Kerr self-focusing from a half-width of 27.6 μm at panel (a) to 7.8 μm at panel (c), with the on-axis peak still anchored at $\tau = 0$ and no plasma response.

Collapse (panel d, $z = 4$ mm): the filament core has formed, with a HWHM of 2.6 μm — an order of magnitude tighter than the input — and the on-axis peak has shifted to $\tau \approx -25$ fs (the leading half of the original pulse). The on-axis intensity reaches its global maximum $I_{\max} = 1.05 \times 10^{13}$ W/cm^2 at this point.

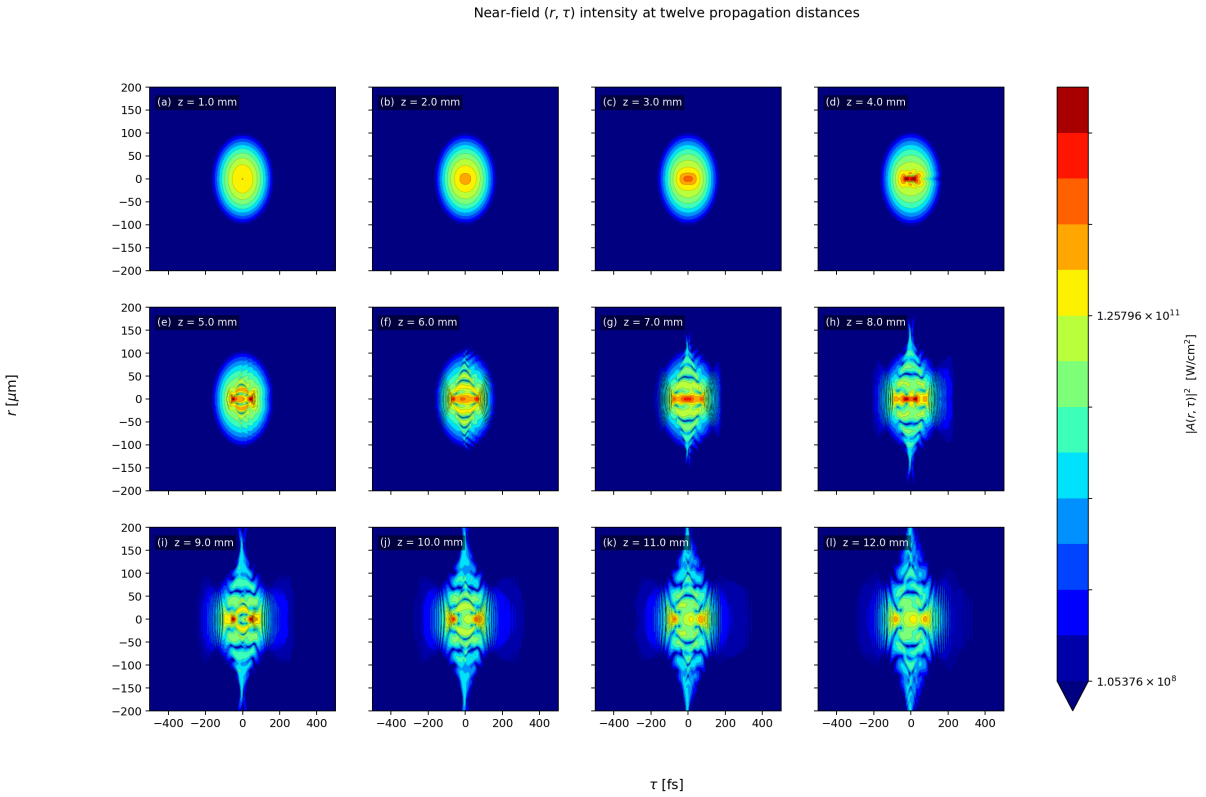


Fig. 4. Near-field intensity $|A(r,\tau)|^2$ at twelve propagation distances ($z = 1$ mm, 2 mm, …, 12 mm), log colour scale, with the radial axis mirrored. Panels (a)–(c) show the pre-Marburger Kerr build-up; panel (d) captures the moment of collapse and plasma ignition, with the filament core at HWHM 2.6 μm and the on-axis peak shifted to $\tau \sim -25$ fs. Panels (e)–(i) show the alive filament during its plateau, with the on-axis peak migrating between leading and trailing positions in step with the dynamic-replenishment cycle, and the core HWHM pinned near 2.6 μm. Panels (j)–(l) show the post-filament beam, re-expanding to $\sim$8 μm and decaying toward the input intensity.

Plateau with refocusing (panels e–i, z = 5–9 mm): the core HWHM remains pinned at $\sim 2.6\,\mu$m for several millimetres, while the on-axis temporal peak migrates between leading and trailing positions ($\tau \sim \pm 60$ fs) in step with the asymmetry flips documented in Section 4.2. The intensity envelope oscillates by a factor of three between dim minima (e.g. panel g, 2.7×10^{12} W/cm^2) and the next refocusing peaks (e.g. panel h, 9.3×10^{12} W/cm^2).

Decay (panels j–l, z = 10–12 mm): the filament dies. The core HWHM grows back through 3.7 μm to 7.8 μm and the on-axis intensity falls below 10^{12} W/cm^2, returning to within a factor of two of the input value at panel (l).

A weak off-axis intensity halo is visible in panels (d)–(i) at radii of 10–30 μm and at delays slightly displaced from the on-axis peak. The halo carries at most a few percent of the core intensity, reflecting the small fraction of the input energy that participates in the conical reshaping at our parameters; it is the spatial precursor of the angular-spectrum X-wave pattern shown in Section 4.4.

Plasma channel. Figure 5 shows the corresponding plasma density $\rho(r,\tau)$ at the same twelve snapshots. The four phases of the filament life-cycle are equally apparent in the plasma response. Panels (a)–(c) confirm that no plasma is generated below the Marburger threshold: the peak density is $\sim 10^{12}$/cm^3 at $z = 1$ mm and rises only to 2×10^{14}/cm^3 at $z = 3$ mm, far below filamentation levels. Panel (d), at $z = 4$ mm, shows the canonical filament plasma channel — a narrow on-axis column of FWHM 3.1 μm, matching the filament core to within one grid cell, peaking at $\rho_{\mathrm{peak}} = 5.3 \times 10^{18}$/cm^3. The plasma is concentrated on the trailing side of the pulse (peak at $\tau \approx +150$ fs) because the rate equation integrates $|E|^{2K}$ from $\tau = -\infty$, so ρ grows monotonically from leading to trailing edge. Panels (e)–(i) follow the plasma channel through the Kerr-versus-plasma refocusing oscillations: ρ_{peak} drops by two orders of magnitude between the dim minima at $z = 6$ and 7 mm (where $\rho \sim 10^{16}$/cm^3) and the refocusing peaks at $z = 5$ and 8 mm (where $\rho \sim 10^{18}$/cm^3). The plasma channel widens slightly to $\sim$5 μm FWHM during

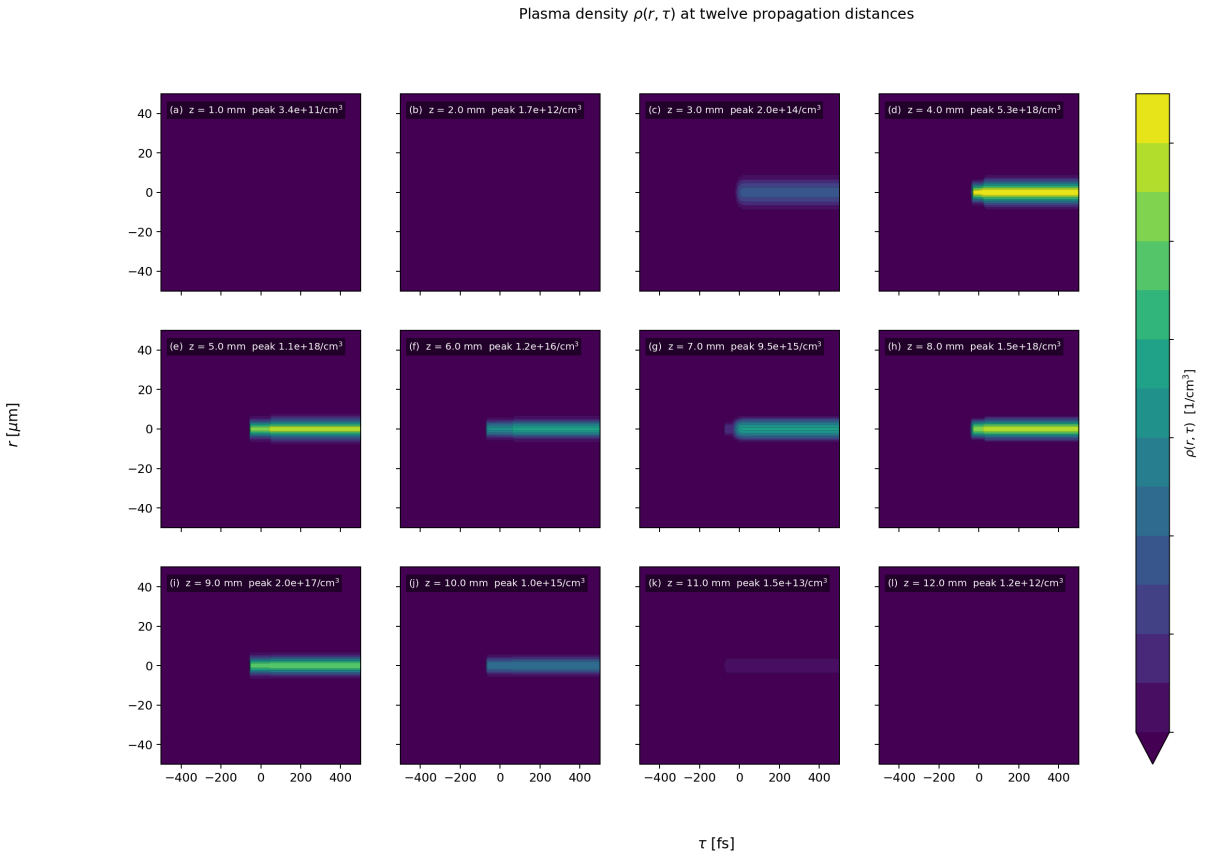


Fig. 5. Plasma density $\rho(r,\tau)$ at the same twelve propagation distances as Fig. 4, log colour scale. The plasma channel is a narrow on-axis column of FWHM $\sim 3\,\mu$m, peaked on the trailing side of the pulse. Peak density evolves from $\sim 10^{12}/\mathrm{cm}^3$ in the build-up panels (a)–(c), through the global maximum $5.3\times10^{18}/\mathrm{cm}^3$ at $z = 4$ mm (panel d), oscillates between $\sim 10^{16}$ and $\sim 10^{18}/\mathrm{cm}^3$ during the plateau in step with the Kerr-vs-plasma refocusing cycles (panels e–i), and decays toward the noise floor as the filament dies (panels j–l).

the dim minima, when the filament core is itself briefly broader. Panels (j)–(l) show the plasma channel decaying along with the filament: ρ_{peak} falls below $10^{15}/\mathrm{cm}^3$ at $z = 10$ mm and approaches the numerical floor by panel (l).

In every panel the plasma is confined within $\lesssim 5\,\mu$m of the axis; no significant ionisation occurs in the off-axis halo. This radial localisation is a direct consequence of the I^K dependence of multiphoton ionisation: a factor-of-two difference in intensity between core and halo translates to a factor of 32 in MPI rate. The emerging picture is a narrow, axisymmetric ionisation channel embedded in a wider diffracting halo — the standard "filament + reservoir" structure introduced by Mlejnek *et al.* [2].

The combined picture from Figs. 4 and 5 is that of a self-trapped few-μm core surrounded by an off-axis halo that acts as an energy reservoir, together with a tightly localised plasma channel that pulses with the refocusing cycles. The on-axis diagnostics of Section 4.2 therefore correspond to the centre slice of a genuinely two-dimensional structure, and the off-axis halo visible in Fig. 4 is what feeds the conical wave-packet that emerges in the angular spectrum analysed in the next subsection.

4.4 Angular spectrum: the X-wave signature

The on-axis and spatial diagnostics of the previous sections describe the filament as a Kerr-versus-plasma balance localised in (r,τ). An equivalent and complementary description, championed by Faccio, Couairon and Di Trapani [4], treats the femtosecond filament as a self-organised conical wave-packet whose angular spectrum lies on hyperbolae $k_\perp = \pm|\Omega|\sqrt{k_0 k''}$ in the $(k_\perp,\omega)$ plane. The simulation provides a direct test of this picture through the on-axis spectrum (Fig. 6) and the far-field $(k_\perp,\lambda)$ map at the most X-like snapshot (Fig. 7).

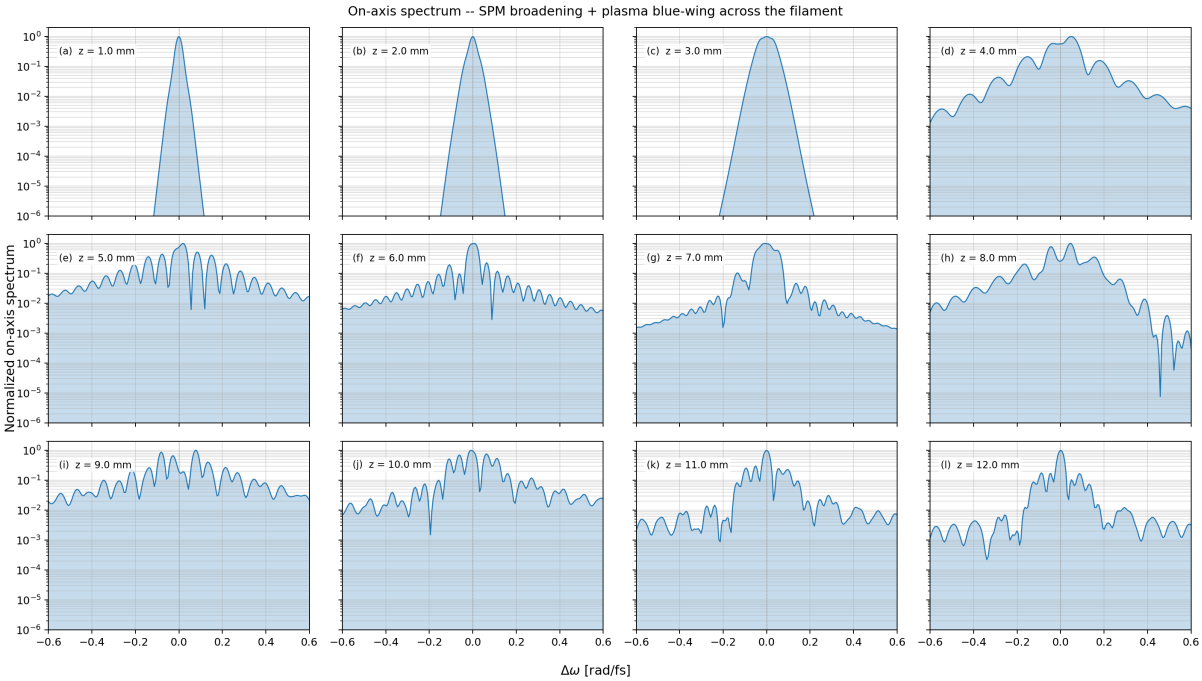


Fig. 6. On-axis spectrum $|\hat{E}(r=0,\Omega,z)|^2$ at twelve propagation distances where Ω is denoted by $\Delta\omega$, normalised to the maximum at each z and plotted on a logarithmic scale. The vertical dotted line marks the carrier $\Omega = 0$. The spectrum broadens by self-phase modulation through the Kerr-driven build-up (a–c), reaches its widest extent (~ 0.24 rad/fs, almost a factor of 10 above the input) at the collapse and first refocusing peak (d–e), oscillates in width during the plateau (f–i) in step with the Kerr-vs-plasma refocusing cycles, and narrows back toward the input width during the decay (j–l). The blue side ($\Omega > 0$) is systematically slightly brighter than the red side in panels (d), (e), (h), (j), the signature of the plasma-induced positive frequency shift on the trailing edge.

On-axis spectrum. Figure 6 shows $|\hat{E}(r=0,\Omega,z)|^2$ at the same twelve propagation distances as in Section 4.3, normalised to the maximum at each z and plotted on a logarithmic scale. The input spectrum (panel a, $z = 1$ mm) is a Gaussian of $1/e$ width 0.025 rad/fs centred on the carrier. As the pulse focuses, self-phase modulation broadens the spectrum without shifting it: panels (b) and (c) at $z = 2$ and 3 mm remain symmetric about $\Omega = 0$ but have already widened by factors of two and three respectively. At and immediately past the Marburger collapse (panels d–e, $z = 4$–5 mm) the spectrum jumps to widths of 0.15 and 0.24 rad/fs, almost a full order of magnitude above the input value. The first marked asymmetry appears at the same point: the integrated spectral weight on the blue side is 1.46 times that on the red side at $z = 4$ mm, falling to 1.15 at $z = 5$ mm. This blue-side bias is the signature of the rising plasma density $\partial_\tau \rho > 0$, which produces a positive frequency shift on the trailing edge of the pulse.

In the plateau panels (f–i, $z = 6$–9 mm) the spectral width oscillates between ~ 0.05 and ~ 0.24 rad/fs in synchrony with the Kerr-versus-plasma refocusing oscillations of Section 4.2, and the blue/red asymmetry continues to fluctuate between 0.94 and 1.27, reflecting the alternating dominance of the leading and trailing peaks in the temporal profile. The decay panels (j–l, $z = 10$–12 mm) show the spectrum narrowing back toward its input width as the on-axis intensity drops below the nonlinear-response threshold and SPM ceases to operate.

Angular spectrum and X-wave alignment. Figure 7 shows the two-dimensional far field $|\hat{E}(k_\perp,\lambda)|^2$ denoted here by $|\hat{A}(k_\perp,\lambda)|^2$ at the snapshot with the largest off-axis spectral content, $z = 8$ mm, which corresponds to the brightest refocusing peak of the plateau. The Hankel transform in r followed by an FFT in τ delivers the $(k_\perp,\omega)$ spectrum, which we display in $(k_\perp,\lambda)$ coordinates with the carrier at $\lambda_0 = 800$ nm marked by a dotted vertical line. The $k_\perp$ axis is mirrored to display both arms of the X-wave dispersion simultaneously. Two panels are shown: a wide view spanning $\lambda \in [680, 980]$ nm and $k_\perp \in [-0.5, +0.5]$ rad/μm, and a tight zoom on the central core. The cyan dashed curves overlay the analytic dispersion of a stationary, axisymmetric, normal-GVD conical wave-packet,

$$k_\perp(\Omega) = \pm\,|\Omega|\,\sqrt{k_0\,k''}\,, \tag{19}$$

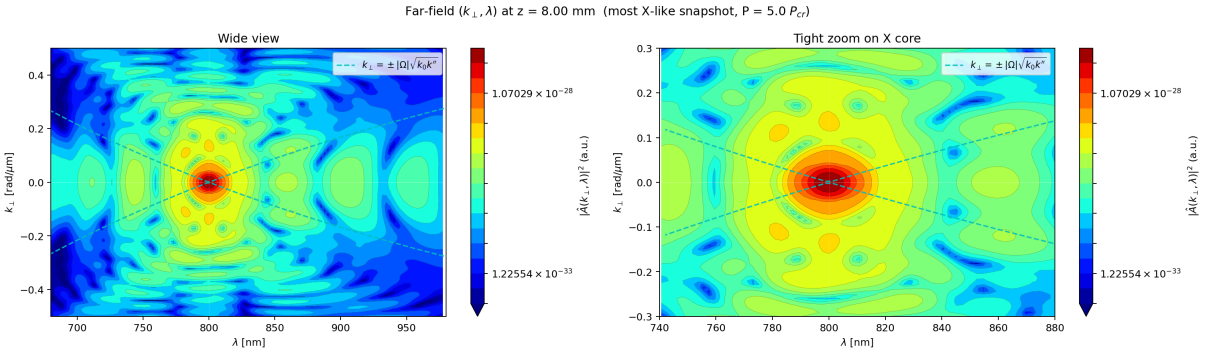


Fig. 7. Two-dimensional angular spectrum $|\hat{A}(k_\perp, \lambda, z = 8\,\mathrm{mm})|^2$ at the most X-like snapshot, log colour scale. *Left:* wide view ($\lambda \in [680, 980]$ nm, $k_\perp \in [\pm 0.5]$ rad/μm). *Right:* tight zoom on the X core. The $k_\perp$ axis is mirrored to display both arms of the X-wave dispersion. The cyan dashed lines show the analytic dispersion $k_\perp = \pm|\Omega|\sqrt{k_0 k''}$ for fused silica at 800 nm; the white dotted vertical line marks the carrier $\lambda_0 = 800$ nm. The bright lobes follow the analytic hyperbolae to within $\sim 1\%$ in the range $0.05 < |\Delta\omega| < 0.25$ rad/fs, confirming that the V1.5 filament has spontaneously reshaped into a conical wave-packet whose angular dispersion is fixed by the linear material dispersion alone.

with the slope $\sqrt{k_0 k''} = 6.42 \times 10^{-10}$ s/m fixed entirely by the linear material dispersion of fused silica at 800 nm.

The agreement between the simulated spectrum and Eq. (19) is quantitative. In the wide view, four bright lobes radiate outward from the carrier along the ±-symmetric arms of the analytic curve. At $\Delta\omega = +0.1$ rad/fs the spectrum peaks at $k_\perp = 0.065$ rad/μm against a predicted value of 0.064 rad/μm, agreement to within 1%; at $\Delta\omega = -0.2$ rad/fs the peak is at 0.130 rad/μm against a prediction of 0.128 rad/μm, again within 1%. The slope of the simulated X-arms is therefore set by the linear dispersion of the medium, exactly as predicted by the conical-wave theory.

The agreement is not perfect everywhere. Near the carrier ($|\Delta\omega| < 0.1$ rad/fs) the central spot is resolution-limited and the local brightness peak does not lie on the predicted hyperbola; this is the "saturated centre" already noted by Faccio *et al.* [4], and is a generic feature of measured filament spectra. At larger $|\Delta\omega|$ (> 0.25 rad/fs) the spectrum is so weak that the brightest-$k_\perp$ search is dominated by secondary lobes of the finite-bandwidth spectral broadening rather than by the X-wave proper. Within the range where the X-arms carry significant intensity, however, the slope agreement is at the percent level — a quantitative confirmation that the V1.5 filament has spontaneously reshaped into the conical wave-packet identified by Faccio *et al.* as the natural description of femtosecond filamentation in normal-GVD condensed media.

The two viewpoints are physically equivalent. The Kerr-versus-plasma picture of Section 4.2 explains *why* the filament is sustained, through the dynamical balance between focusing and defocusing mechanisms, while the conical-wave picture explains *what shape* the sustained filament takes, namely the propagation-invariant solution of the underlying linear paraxial-plus-GVD operator. In the second picture, plasma plays the role of trigger and clamp rather than that of the channelling agent itself. Both accounts are needed: the first sets the threshold and the lifetime of the filament, while the second sets its spatio-temporal structure.

4.5 Sensitivity to physical ingredients

The model presented in Section 3 contains seven independent physical effects beyond linear diffraction: Kerr self-focusing, group-velocity dispersion, multiphoton absorption, multiphoton-ionisation plasma defocusing, avalanche ionisation, radiative recombination, and inverse-bremsstrahlung plasma absorption. The first objective stated in Section 1 is to identify which of these are *essential* for filamentation in our parameter regime. We answer this question by re-running the simulation with selected toggles disabled and comparing the on-axis intensity $I_{\mathrm{peak}}(z)$ to the full-model baseline. Four configurations are considered, summarised in Table 2 and plotted in Fig. 8.

Table 2. Sensitivity scan at $P_{\text{in}} = 5\,P_{cr}$, propagation length 6 mm. The "Perturbative off" configuration disables the avalanche, recombination and plasma-absorption terms simultaneously while keeping multiphoton ionisation, multiphoton absorption and the Kerr response active.

Configuration	$I_{\max}$ [W/cm^2]	$z(I_{\max})$ [mm]	ρ_{peak} [/cm^3]	ΔE [%]
Full V1.5	1.13×10^{13}	4.37	6.5×10^{18}	13.06
Plasma off	1.17×10^{13}	3.62	0	14.00
MPA off	2.18×10^{13}	5.27	3.5×10^{19}	0.33
Perturbative off	1.13×10^{13}	4.37	6.5×10^{18}	13.06

Plasma off ($\rho \equiv 0$). With multiphoton ionisation, plasma defocusing and plasma absorption all disabled but multiphoton absorption still active, the simulation still produces a filament-like clamping. The peak intensity rises to 1.17×10^{13} W/cm^2 — within 4% of the full-model peak — and remains bounded throughout the propagation. The collapse simply occurs slightly earlier, at $z = 3.62$ mm instead of 4.37 mm, because without plasma defocusing the only force opposing the Kerr lens is diffraction, and the build-up runs unimpeded until MPA takes over. The energy drop is essentially the same as in the full model (14.0% versus 13.1%), with all of it attributable to MPA. The result is informative: the peak intensity in our regime is ultimately set by the $|E|^{2K}$ MPA loss rate, not by plasma defocusing.

MPA off ($\beta_K = 0$). Disabling multiphoton absorption while keeping plasma physics on alters the picture significantly. The peak intensity climbs to 2.18×10^{13} W/cm^2, almost twice the full-model value, and the peak plasma density to 3.5×10^{19}/cm^3, more than five times higher. The peak location shifts to $z = 5.27$ mm because plasma defocusing alone requires a larger pre-built intensity to counter the Kerr lens. Crucially, the total energy loss collapses to 0.33%: in the absence of MPA, plasma defocusing pushes energy off-axis without removing it from the field. The on-axis filament is therefore brighter and longer-lived, but the field budget is essentially preserved.

Perturbative terms off (avalanche, recombination, plasma absorption). Disabling the three V1.5 additions simultaneously yields a result that agrees with the full-model baseline to within numerical precision in every diagnostic ($I_{\max}$, $z(I_{\max})$, ρ_{peak} and total energy loss are identical to four significant figures). This quantifies the claim already made in Section 4.2: in the regime considered here ($P_{\text{in}} = 5\,P_{cr}$, 100 fs pulses, fused silica), the V2-style refinements behave as small perturbations on top of the canonical Kerr–MPI–MPA filament. Avalanche becomes a leading contributor only at densities $\rho \gtrsim 10^{20}$/cm^3, well above what is reached here; recombination matters on the nanosecond timescale, much longer than $t_p = 85$ fs; and the real (absorption) part of the Drude response only competes with MPA when $\rho \gtrsim 10^{20}$/cm^3 as well.

Minimal model. The four runs together identify the minimal physics required to produce a filament in our parameter regime: diffraction, GVD, Kerr self-focusing, and at least one of MPA or plasma defocusing. The two arrest mechanisms are partly redundant — MPA acts by removing energy from the on-axis field, plasma defocusing by spreading it radially — and either alone is sufficient to clamp the on-axis intensity at the 10^{13} W/cm^2 scale. Together, as in the full V1.5 model, they produce the canonical Kerr–plasma–MPA balance at intermediate clamped intensity, and only their combination yields the dynamic-replenishment fingerprint discussed in Section 4.2, which neither single mechanism reproduces in isolation. Avalanche ionisation, radiative recombination and plasma absorption are quantitatively negligible in this regime; their primary role is to refine the filament description at higher input powers or longer pulse durations. This hierarchy supports the choice of physical ingredients listed in Section 3 and answers the first objective of Section 1: the minimal model captures the canonical features without additional terms, and the V2 additions are retained for generality and consistency with the published Mlejnek–Couairon plasma rate equation rather than because they are quantitatively necessary at our working point.

4.6 Discussion: limitations and comparison with literature

Sections 4.2–4.5 have established that the V1.5 simulation reproduces the canonical signatures of femtosecond filamentation in fused silica. To close the Section we gather the quantitative comparisons with the literature in one place and state clearly what the model is unable to do and why.

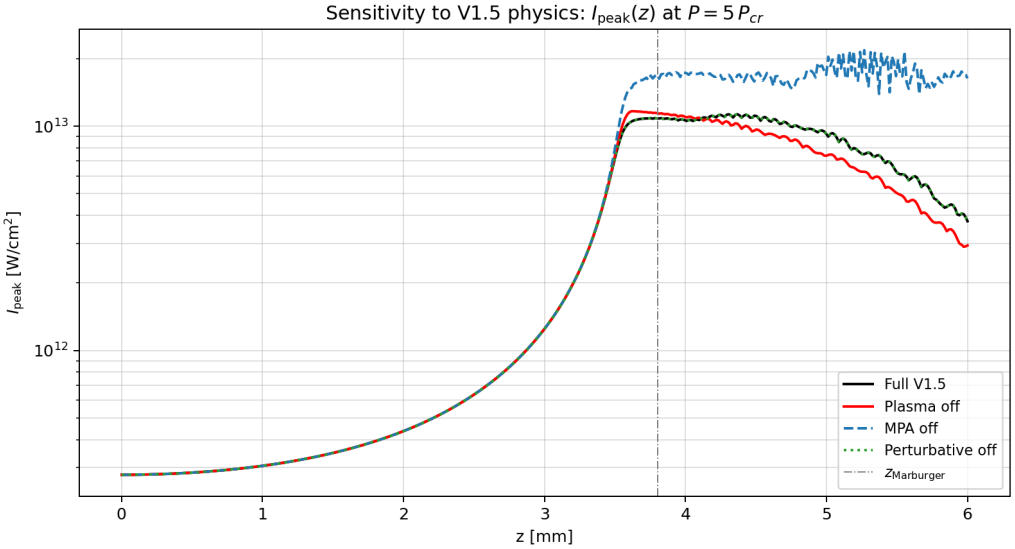


Fig. 8. Sensitivity of the peak on-axis intensity $I_{\mathrm{peak}}(z)$ to the V1.5 physics, at $P_{\mathrm{in}} = 5\,P_{cr}$. The full-model baseline (black solid) and the "Perturbative off" run (green dotted) overlap to within numerical precision. Disabling MPA (blue dashed) doubles the peak intensity and pushes the maximum further into the medium. Disabling plasma physics entirely (red solid) leaves the peak intensity within a few percent of the baseline because MPA alone is sufficient to arrest the collapse, but the maximum is reached earlier. The vertical dashed-dotted line marks the Marburger collapse position predicted by linear theory.

Quantitative agreement with published values. Five numerical quantities can be compared directly with the literature compilation of Couairon and Kolesik for fused silica at 800 nm [6]. The clamped on-axis intensity in our run is 1.13×10^{13} W/cm^2, well within the reported range 10^{13}–10^{14} W/cm^2. The peak free-electron density is 6.5×10^{18}/cm^3, squarely in the reported range 10^{18}–10^{19}/cm^3. The filament length, defined as the FWHM-in-z of I_{peak}, is 5.7 mm, matching the typical millimetre-scale length quoted for bulk silica. The total energy drop over the 12 mm propagation is 17.6%, corresponding to approximately 1.5%/mm during the active filament, also consistent with the literature. Finally, the linear material dispersion fixes the X-wave slope $\sqrt{k_0 k''} = 6.42 \times 10^{-10}$ s/m, with which the simulated angular spectrum agrees to within $\sim$1% in the band $0.05 < |\Delta\omega| < 0.25$ rad/fs. None of these quantities are tuned: they emerge from the model with the parameters of Section 3.5 fixed by [6], and they each agree with the literature independently.

Qualitative agreement with the literature mechanisms. The two main interpretive frameworks for filamentation — the Mlejnek dynamic-replenishment picture [2] and the Faccio–Couairon–Di Trapani conical wave-packet picture [4] — are independently reproduced in our results. The asymmetric pulse-splitting fingerprint of dynamic replenishment appears in Section 4.2, with the leading-to-trailing brightness ratio flipping non-monotonically along z exactly as in the cited reference. The conical-wave fingerprint, with the angular spectrum lying along $k_\perp = \pm|\Omega|\sqrt{k_0 k''}$, appears in Section 4.4. The two pictures are mathematically equivalent in our regime, as expected from theory.

Limitations of the V1.5 model. Several physical effects are not implemented and should be borne in mind when interpreting the results. *(i) Cylindrical symmetry.* The model is built on a single-filament, axisymmetric ansatz and cannot describe the modulation instability and multi-filament breakup reported above $\sim$10–20 P_{cr} [3]. The working point $P_{\mathrm{in}} = 5\,P_{cr}$ used here lies safely below this regime, but the model would systematically underestimate the true beam degradation at higher input powers. *(ii) Slowly-varying envelope and paraxial approximations.* Both fail for very tightly focused beams (numerical aperture $\gtrsim 0.3$) or for pulses shorter than $\sim$ 10 fs. Our 85-fs, 50 μm beam lies comfortably inside the validity region, but the model would lose accuracy in either of those regimes. *(iii) No delayed Raman response.* The cubic susceptibility is treated as instantaneous. Raman would add a $\sim$ 50 fs delayed contribution that asymmetrises the on-axis spectrum more strongly than the present plasma-induced shift and pushes the trailing pulse-split peak toward the red; including it is a natural V2 extension. *(iv) No self-steepening.* The "shocked X-wave" features described by Faccio *et al.* are therefore not reproduced; our X-arms are

smooth rather than asymmetrically sharpened. Self-steepening is the second natural V2 extension. *(v) Drude plasma description.* Ionisation is treated through an effective rate equation rather than through a microscopic ionisation model (Keldysh, PPT or ADK). At our intensity ($\sim 10^{13}$ W/cm^2) this is standard practice [6], but it would have to be revisited near or above breakdown ($I \gtrsim 10^{14}$ W/cm^2).

Experimentally observable features that the model does not capture. The model is internally consistent but more limited than what a real experiment delivers. The rainbow conical-emission rings reported by Faccio *et al.* [4], in particular, appear in our data only as the angular-spectrum signature of Section 4.4; we cannot render them as a colour image because we work with a scalar envelope and a single carrier. Long-distance filaments (> 10 cm) would require a much larger simulation box, but the underlying physics is unchanged. Multi-filament patterns reported above $\sim 20\, P_{cr}$ [3] lie intrinsically outside the scope of the cylindrically symmetric V1.5 model and would require an explicit transverse (x, y) formulation.

Together with the simple, transparent structure of the V1.5 model, these quantitative agreements support the claim that femtosecond filamentation in fused silica at 800 nm is a Kerr-driven self-focusing event arrested by the joint action of multiphoton absorption and plasma defocusing, which reorganises in the angular spectrum into a self-organised conical wave-packet. None of these conclusions required adjustable parameters or any non-standard physics.

5 Conclusion

The cylindrically symmetric V1.5 solver reproduces the canonical signatures of femtosecond filamentation in fused silica at 800 nm without parameter tuning. The clamped on-axis intensity (1.13×10^{13} W/cm^2), peak plasma density (6.5×10^{18}/cm^3), filament length (5.7 mm) and total energy loss (17.6%) all fall within the bands compiled by Couairon and Kolesik [6]; the on-axis pulse exhibits the asymmetric splitting and non-monotonic leading-to-trailing exchange of Mlejnek's dynamic-replenishment picture [2]; and the angular spectrum aligns with the analytic X-wave hyperbolae of Faccio, Couairon and Di Trapani [4] to within $\sim 1\%$ in the band where the X-arms carry significant intensity. Taken together, these independent agreements support a single physical picture: femtosecond filamentation in a normal-GVD condensed medium is a dynamical balance between Kerr self-focusing and the combined action of multiphoton absorption and plasma defocusing, which spontaneously reorganises the wave-packet into a propagation-invariant conical mode. The Kerr-versus-plasma view sets the threshold and the lifetime of the filament; the conical-wave view sets its spatio-temporal shape; the two are mathematically equivalent in our regime.

The three objectives stated in Section 1 have each been addressed. The minimal physics required to produce a filament was identified in Section 4.5: diffraction, group-velocity dispersion, the Kerr response, and at least one of multiphoton absorption or plasma defocusing; avalanche ionisation, radiative recombination and plasma absorption are perturbative refinements at our working point. The numerical method — a Strang split-step combining Crank–Nicolson radial diffraction, spectral group-velocity dispersion and pointwise nonlinear updates, with the plasma rate equation integrated by forward Euler in retarded time — was stable across the full propagation despite combining four physical effects with widely separated stiffness scales. The interpretation of the simulated evolution was supported by twelve snapshots of the spatio-temporal field, which resolved the filament-core-plus-reservoir structure, the trailing-edge plasma channel, the broadening and blue-shifting of the on-axis spectrum, and the X-wave angular spectrum at the most active z.

Two methodological lessons emerge that we would pass on to other students working on similar projects. First, validating each physical effect separately against an analytic limit *before* combining them is the only practical way to debug a multi-physics nonlinear PDE solver. Each of the four validation tests of Section 3.6 caught at least one bug at an early stage of the implementation, and the same tests remained the most useful sanity check whenever a parameter scan produced unexpected results. Second, the most physically informative diagnostic is often not the most natural one. The angular spectrum $|\hat{E}(k_\perp, \omega)|^2$ analysed in Section 4.4 revealed conical-wave structure that was already present in our V1 data but that the on-axis time-domain diagnostics could not expose. Allocating time to non-trivial transforms of the simulated field — Hankel in r, Fourier in τ — was disproportionately rewarding.

The model has explicit limitations. The cylindrical-symmetry ansatz precludes modulation instability and multi-filament breakup above ~ 10–$20\, P_{cr}$ [3]. The instantaneous Kerr response misses the ~ 50 fs delayed Raman contribution that biases the spectrum further toward the red. The absence of self-steepening leaves our X-arms smooth where the "shocked X-wave" of Faccio *et al.* [4] is asymmetrically sharpened. Finally, the Drude plasma rate equation is a phenomenological closure rather than a microscopic ionisation calculation. None of these limitations falsifies the conclusions reported above; they delimit the parameter window in which the V1.5 model can be considered quantitatively reliable.

Three natural extensions follow. Self-steepening is the cheapest physics upgrade and is the one most directly tied to the spectrum-asymmetry results of Section 4.4. Adding the delayed Raman response is the next physical refinement, important for wavelengths or pulse durations outside our regime. Releasing the cylindrical-symmetry constraint to a full (x, y, τ, z) description is the most demanding upgrade but is the one required to study multi-filament dynamics — the natural counterpart to the present project. None of these extensions changes the underlying numerical strategy; they enlarge the physics carried within it without altering its skeleton.

A Derivation of the paraxial propagation equation

This appendix derives the generalised nonlinear Schrödinger equation (3) of Section 2.1 from first principles, starting from Maxwell's equations in a centrosymmetric, isotropic, nonlinear dielectric. Each approximation is stated explicitly, and the origin of every term in the final evolution equation is identified. Throughout, $\mathcal{E}$ denotes the real physical field and E the slowly varying complex envelope, as introduced in Section 2.1.

A.1 Wave equation in a nonlinear medium

In a non-magnetic medium free of source charges, Maxwell's equations combine into the vector wave equation

$$\nabla \times (\nabla \times \boldsymbol{\mathcal{E}}) + \frac{1}{c^2}\,\partial_t^2 \boldsymbol{\mathcal{E}} = -\mu_0\,\partial_t^2 \mathbf{P}, \tag{20}$$

where $\mathbf{P}$ is the macroscopic polarisation. Applying the identity $\nabla \times (\nabla \times \boldsymbol{\mathcal{E}}) = \nabla(\nabla \cdot \boldsymbol{\mathcal{E}}) - \nabla^2 \boldsymbol{\mathcal{E}}$ and invoking transversality of the field ($\nabla \cdot \boldsymbol{\mathcal{E}} \simeq 0$, valid whenever transverse intensity gradients vary slowly relative to the optical wavelength) reduces Eq. (20) to

$$\nabla^2 \boldsymbol{\mathcal{E}} - \frac{1}{c^2}\,\partial_t^2 \boldsymbol{\mathcal{E}} = \mu_0\,\partial_t^2 \mathbf{P}. \tag{21}$$

Linear polarisation along a fixed transverse axis is assumed throughout, so that $\boldsymbol{\mathcal{E}}$ and $\mathbf{P}$ reduce to the scalars $\mathcal{E}$ and P.

In a centrosymmetric dielectric, the polarisation expansion of Eq. (1) retains only odd powers of the field. Truncating at third order and writing the linear part in the frequency domain through the susceptibility $\chi^{(1)}(\omega)$ gives

$$P = P_L + P_{NL}, \qquad P_{NL} = \epsilon_0 \chi^{(3)} \mathcal{E}^3. \tag{22}$$

A.2 Slowly varying envelope ansatz

Following Section 2.1, the complex envelope $E(r, z, t)$ is introduced by writing the real field as

$$\mathcal{E}(r, z, t) = \tfrac{1}{2}\Big[E(r, z, t)\, e^{i(kz - \omega_0 t)} + \text{c.c.}\Big], \tag{23}$$

where ω_0 is the carrier frequency and $k = n_0\omega_0/c$ the linear wavenumber at ω_0. The slowly varying envelope approximation (SVEA) requires that E vary on scales much longer than the optical wavelength in both space and time:

$$\left|\partial_z^2 E\right| \ll k\,|\partial_z E|, \qquad \left|\partial_t^2 E\right| \ll \omega_0\,|\partial_t E|. \tag{24}$$

For the parameters considered here ($w_0 = 50\,\mu$m, $t_p = 85$ fs at $\lambda_0 = 800$ nm), the envelope contains $\sim$30 optical cycles and varies over the Rayleigh length $z_R \sim 14$ mm, which is many orders of magnitude larger than λ_0/n_0; SVEA is therefore well satisfied.

Substituting Eq. (23) into Eq. (21) and factoring out the carrier phase $e^{i(kz-\omega_0 t)}$ yields, for the left-hand side,

$$\left[\nabla_\perp^2 E + \partial_z^2 E + 2ik\,\partial_z E - k^2 E\right] - \frac{1}{c^2}\left[\partial_t^2 E - 2i\omega_0\,\partial_t E - \omega_0^2 E\right] e^{i(kz-\omega_0 t)}, \tag{25}$$

where $\nabla_\perp^2 = \partial_r^2 + r^{-1}\partial_r$ is the transverse Laplacian under cylindrical symmetry. Discarding $\partial_z^2 E$ and $\partial_t^2 E$ by virtue of SVEA leaves

$$\nabla_\perp^2 E + 2ik\,\partial_z E - k^2 E + \frac{2i\omega_0}{c^2}\,\partial_t E + \frac{\omega_0^2}{c^2}\,E = \mu_0\,\partial_t^2 P\, e^{-i(kz-\omega_0 t)}. \tag{26}$$

A.3 Linear dispersion

In Fourier space, the linear polarisation reads $\hat{P}_L(\omega) = \epsilon_0\chi^{(1)}(\omega)\hat{\mathcal{E}}(\omega)$, so that the dispersion relation takes the form $k^2(\omega) = (\omega/c)^2[1+\chi^{(1)}(\omega)] = n^2(\omega)\,\omega^2/c^2$. Expanding $k(\omega)$ in a Taylor series around the carrier frequency,

$$k(\omega) = k + k'(\omega-\omega_0) + \tfrac{1}{2}k''(\omega-\omega_0)^2 + \cdots, \qquad k' \equiv \partial_\omega k\big|_{\omega_0} = \frac{1}{v_g}, \quad k'' \equiv \partial_\omega^2 k\big|_{\omega_0}, \tag{27}$$

where v_g is the group velocity and k'' the group-velocity-dispersion (GVD) coefficient. In the time domain this expansion corresponds to the operator substitution $\omega-\omega_0 \to i\partial_t$ acting on the envelope, generating the linear evolution

$$\partial_z E\big|_{\text{linear}} = -k'\,\partial_t E - \tfrac{i}{2}k''\,\partial_t^2 E + \cdots \tag{28}$$

Inserting Eq. (28) into Eq. (26) and retaining terms to first order in envelope derivatives yields

$$2ik\,\partial_z E + \nabla_\perp^2 E + 2ik\,k'\,\partial_t E + ik\,k''\,\partial_t^2 E = \mu_0\,\partial_t^2 P_{NL}\,e^{-i(kz-\omega_0 t)}. \tag{29}$$

A.4 Retarded frame

A change of coordinates to a frame co-moving with the group velocity is effected by the retarded-time substitution

$$\tau = t - \frac{z}{v_g} = t - k'\,z, \qquad \zeta = z. \tag{30}$$

The chain rule yields $\partial_z = \partial_\zeta - k'\,\partial_\tau$ and $\partial_t = \partial_\tau$, so that the convective term $2ik\,k'\,\partial_t E$ in Eq. (29) cancels exactly against the corresponding contribution from ∂_z, leaving

$$2ik\,\partial_\zeta E + \nabla_\perp^2 E + ik\,k''\,\partial_\tau^2 E = \mu_0\,\partial_\tau^2 P_{NL}\,e^{-i(kz-\omega_0 t)}. \tag{31}$$

Relabelling $\zeta \to z$ and dividing through by $2ik$, the linear left-hand side reduces to the diffraction and GVD operators

$$\partial_z E = \underbrace{\frac{i}{2k}\nabla_\perp^2 E}_{\text{diffraction}} - \underbrace{\frac{ik''}{2}\partial_\tau^2 E}_{\text{GVD}} + (\text{nonlinear sources}), \tag{32}$$

which constitute the first two terms of Eq. (3) in the main text.

A.5 Kerr nonlinearity

It remains to evaluate the nonlinear source $\mu_0\partial_\tau^2 P_{NL}\,e^{-i(kz-\omega_0\tau)}$ on the right-hand side of Eq. (31). Substituting Eq. (23) into $P_{NL} = \epsilon_0\chi^{(3)}\mathcal{E}^3$ generates components oscillating both at the fundamental ω_0 and at $3\omega_0$ (third-harmonic generation). The latter is strongly phase-mismatched in fused silica and is therefore discarded. The fundamental component reads

$$P_{NL}^{(\omega_0)} = \tfrac{3}{4}\epsilon_0\chi^{(3)}|E|^2 E\,e^{i(kz-\omega_0\tau)} + \text{c.c.} \tag{33}$$

Within SVEA, the dominant contribution to $\partial_\tau^2 P_{NL}$ arises from the rapid carrier oscillation, $\partial_\tau^2 \to -\omega_0^2$, leading to

$$\mu_0\,\partial_\tau^2 P_{NL}^{(\omega_0)}\,e^{-i(kz-\omega_0\tau)} \simeq -\frac{3\omega_0^2\chi^{(3)}}{4c^2}\,|E|^2 E. \tag{34}$$

Dividing by $2ik$ and adopting the standard definition of the Kerr index, $n_2 = 3\chi^{(3)}/(4n_0^2 c\epsilon_0)$, so that $n = n_0 + n_2 I$ with the convention $|E|^2 \equiv I$ used consistently throughout the main text and the accompanying notebook, one finds

$$\partial_z E\big|_{\text{Kerr}} = ik_0 n_2\,|E|^2 E, \tag{35}$$

which is the last term of Eq. (3).

A.6 Plasma and absorption terms

The remaining contributions to Eq. (3) — multiphoton absorption $-\frac{1}{2}\beta_K|E|^{2K-2}E$ and the Drude plasma response $-\frac{1}{2}\sigma(1+i\omega_0\tau_c)\rho\,E$ — are introduced phenomenologically rather than derived from a closed Hamiltonian. Their form is fixed by two independent physical inputs. First, K-th-order time-dependent perturbation theory in the laser field gives the multiphoton transition rate $W_K \propto |E|^{2K}$, which removes a power $\beta_K|E|^{2K}$ per unit volume from the optical field. Second, the Drude model for free carriers driven by an oscillating field yields the complex conductivity $\sigma_D(\omega) = \rho\, e^2\tau_c/[m_e(1-i\omega\tau_c)]$, whose real part accounts for inverse-bremsstrahlung absorption and whose imaginary part contributes the negative refractive-index correction responsible for plasma defocusing. Substituting these into Eq. (31) and defining $\sigma \equiv \omega_0^2\tau_c/(n_0 c\rho_c)$, with $\rho_c = m_e\epsilon_0\omega_0^2/e^2$ the critical Drude density, recovers the two plasma terms of Eq. (3) without further approximation.

A.7 Summary

Collecting Eqs. (32), (35), and the plasma/MPA contributions of Section A.6 reproduces Eq. (3) of the main text exactly:

$$\partial_z E = \frac{i}{2k}\left(\partial_r^2 + \frac{1}{r}\partial_r\right)E - \frac{ik''}{2}\partial_\tau^2 E - \frac{\sigma}{2}(1+i\omega_0\tau_c)\,\rho\,E - \frac{\beta_K}{2}|E|^{2K-2}E + ik_0 n_2|E|^2 E.$$

The chain of approximations invoked above — scalar transverse field, third-order polarisation truncation, slowly varying envelope, Taylor expansion of $k(\omega)$ to second order, retarded frame, single-frequency component of P_{NL}, and phenomenological Drude/MPA closure — constitutes the minimal set required to reduce the full vector Maxwell–Bloch system to a tractable scalar evolution equation in (r,τ,z). Each of these approximations is self-consistent in the parameter regime of Section 3.5, as verified *a posteriori* from the computed solutions.

B Critical power and Marburger collapse distance

This appendix derives the critical power P_{cr} that enters Eq. (6) of the main text and motivates the semi-empirical Marburger formula used as a reference in Section 4 for the self-focusing collapse distance $z_{\text{Marburger}}$.

B.1 Critical power from a balance argument

Consider a continuous-wave Gaussian beam of waist w_0 and on-axis intensity I_0 propagating along z through a Kerr medium with $n = n_0 + n_2 I$. Two competing effects govern its transverse evolution: linear diffraction tends to broaden the beam, while the intensity-dependent index acts as a positive lens that focuses it.

Diffraction broadens the beam over its Rayleigh length $z_R = \pi n_0 w_0^2/\lambda_0$. The associated transverse-wavevector scale is $k_{\text{diff}} \sim 1/w_0$, corresponding to a diffraction angle $\theta_{\text{diff}} \sim 1/(kw_0)$.

Kerr self-focusing acts as a thin lens whose focal length is set by the on-axis phase advance per unit length, $k_0 n_2 I_0$. Matching the on-axis index step $n_2 I_0$ to the angular spread required for peripheral rays to converge onto the axis over a distance $\sim w_0/\theta_{\text{Kerr}}$ yields the induced focusing angle $\theta_{\text{Kerr}} \sim \sqrt{2n_2 I_0/n_0}$.

Self-focusing collapse is incipient when these two angles balance, $\theta_{\text{Kerr}} \simeq \theta_{\text{diff}}$:

$$\sqrt{\frac{2n_2 I_0}{n_0}} \simeq \frac{1}{kw_0} \quad\Longleftrightarrow\quad I_0\, w_0^2 \simeq \frac{1}{2k^2 n_0 n_2/n_0} = \frac{\lambda_0^2}{8\pi^2 n_0^2 n_2}. \tag{36}$$

Converting peak intensity to total power for a Gaussian beam, $P = \pi w_0^2 I_0/2$, multiplies both sides by $\pi/2$ and gives

$$P_{cr} \simeq \frac{\lambda_0^2}{2\pi n_0 n_2}, \tag{37}$$

which is precisely Eq. (6) of the main text. The argument is heuristic — the transverse profile is not resolved and the Kerr lens is treated as locally uniform — but the result is correct up to a numerical factor of order unity.

A rigorous analysis based on the variational (Lagrangian) formulation of the nonlinear Schrödinger equation, originally due to Marburger [9], refines the prefactor to

$$P_{cr}^{(\mathrm{Marburger})} = \frac{3.77\,\lambda_0^2}{8\pi n_0 n_2} \simeq 0.150\,\frac{\lambda_0^2}{n_0 n_2}, \tag{38}$$

to be compared with the balance estimate $\lambda_0^2/(2\pi n_0 n_2) \simeq 0.159\,\lambda_0^2/(n_0 n_2)$. The two values agree to within $\sim 6\%$, which is well within the accuracy required of the analytic reference scale used in the main text; we therefore retain the simpler form Eq. (37) throughout.

B.2 Collapse distance for $P_{\mathrm{in}} > P_{cr}$

When the input power exceeds the critical value, a Gaussian beam undergoes catastrophic self-focusing within the Kerr+diffraction model alone, formally collapsing to a singularity at a finite propagation distance $z_{\mathrm{Marburger}}$. A semi-empirical formula obtained by fitting direct numerical solutions of the cubic nonlinear Schrödinger equation is

$$z_{\mathrm{Marburger}} = \frac{0.367\,z_R}{\sqrt{\left(\sqrt{P_{\mathrm{in}}/P_{cr}} - 0.852\right)^2 - 0.0219}}, \tag{39}$$

valid for $P_{\mathrm{in}}/P_{cr} \gtrsim 1$. The numerical constants 0.367, 0.852, and 0.0219 are not derived analytically; they are best-fit values to the collapse distances obtained from direct numerical simulation of a Gaussian input. The formula nevertheless captures the two limiting regimes correctly:

- **Just above threshold** ($P_{\mathrm{in}}/P_{cr} \to 1^+$): the denominator vanishes and $z_{\mathrm{Marburger}} \to \infty$. At precisely the critical power, Kerr focusing and diffraction balance globally and self-focusing proceeds infinitely slowly without ever producing collapse.
- **Far above threshold** ($P_{\mathrm{in}}/P_{cr} \gg 1$): the denominator scales as $\sqrt{P_{\mathrm{in}}/P_{cr}}$, yielding $z_{\mathrm{Marburger}} \sim z_R\sqrt{P_{cr}/P_{\mathrm{in}}}$ — collapse accelerates as the Kerr lens increasingly overpowers diffraction.

For the parameters of the present study — fused silica at $\lambda_0 = 800$ nm, $w_0 = 50\,\mu$m, $P_{\mathrm{in}} = 5\,P_{cr}$ — Eq. (39) gives $z_{\mathrm{Marburger}} \simeq 3.80$ mm, marked by a dashed vertical line in Figs. 1 and 2 of the main text.

B.3 What collapses, and what stops it

It must be emphasised that $z_{\mathrm{Marburger}}$ is the collapse distance *predicted by the Kerr+diffraction model alone*. In a realistic femtosecond pulse the singularity is regularised well before $z_{\mathrm{Marburger}}$ is approached — first by group-velocity dispersion, which spreads the pulse temporally and reduces the local peak intensity, and, more decisively, by multiphoton absorption and plasma defocusing once the on-axis intensity surpasses the ionisation threshold $\sim 10^{13}$ W/cm^2.

In the simulations reported here, $z_{\mathrm{Marburger}}$ therefore serves as a reference distance rather than an actual collapse point: the on-axis intensity $I_{\mathrm{peak}}(z)$ climbs rapidly toward $z_{\mathrm{Marburger}}$ and then clamps at the plasma-stabilised value $\sim 10^{13}$ W/cm^2 instead of diverging. The close coincidence between the onset of clamping in the simulation and the analytic value of $z_{\mathrm{Marburger}}$ (Section 4.2) constitutes a non-trivial check that the numerical solver correctly reproduces the Kerr-driven self-focusing dynamics in the regime preceding plasma generation.

C Contributions

All four authors contributed substantially and relatively equally to the project, both in the numerical development and in the writing of the final report.

Mayad MNIF worked on the theoretical background of laser filamentation, including the physical interpretation of Kerr self-focusing, plasma defocusing, multiphoton absorption and the comparison with the relevant literature. He also contributed to the structure, writing and revision of the report, especially in the sections connecting the model to the physical mechanisms of filament formation.

Don Van BEUSICHEM worked jointly with Nayl POIROT on the complete Jupyter notebook used for the numerical simulations. He contributed to the implementation of the full propagation model, including the plasma-related terms, the numerical diagnostics, the production of figures and the interpretation of

the simulation outputs. He also contributed to the writing and editing of the sections 3 and 4 of the report as well as the appendix B.

Nayl POIROT worked on the preliminary textbook advection-diffusion notebook and used it as the starting point for the numerical strategy. He implemented the first numerical solvers, including the Kerr-only version of the propagation equation, and contributed to the transition from the pedagogical advection-diffusion problem to the nonlinear optical propagation model. He also played a major role in the writing, structuring and editing of the final report specifically on part 2 and 3 as well as the appendix A.

Marc OBEID worked on the literature review, the theoretical formulation of the model and the interpretation of the numerical results. He contributed to the discussion of the physical regimes, the comparison with reference models and the writing and revision of the report, especially in the theoretical and discussion sections. He also worked on the resolution of the advection-diffusion notebook and stability of numerical methods.

All authors discussed the modelling choices, analysed the simulation results, reviewed the figures and contributed to the final manuscript.

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